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Research paper

Reconfigured continuous normalized accelerated gradient flow for computing ground states of spin-1 Bose–Einstein condensates

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ABSTRACT

Based on the mean field Gross–Pitaevskii theory, the ground state of a spin-1 Bose–Einstein condensate (BEC) can be modeled as a vector wave function that minimizes the energy functional of the coupled Gross–Pitaevskii equations, subject to conservation constraints on total mass and magnetization. The normalized gradient flow methods are widely used for computing the ground states of spin-1 BECs due to their simplicity in implementation and efficiency in each time step. However, their slow convergence speed requires many iterations to reach the ground state, which results in excessive computational costs. In this paper, we propose a highly efficient method for computing the ground states of spin-1 BECs, based on the rapid convergence properties of accelerated gradient flow. We establish a reconfigured continuous normalized accelerated gradient flow (RCNAGF) to simulate the spin-1 BEC ground state. One explicit and two semi-implicit schemes are presented for the numerical solution of the RCNAGF, incorporating a Fourier pseudospectral method for spatial discretization. Our proposed schemes are efficient at each time step because the explicit scheme can be directly updated, while the semi-implicit schemes only need to solve several linear systems with constant coefficients. All schemes require solving an additional nonlinear algebraic system, but this computational cost is considered negligible. The numerical experimental results indicate that our schemes are more efficient than those based on normalized gradient flow schemes for computing the ground states of spin-1 BECs. Specifically, our explicit scheme allows for larger stable time step sizes, while the semi-implicit schemes significantly reduce the number of iterations required to meet the stopping criterion at the same step size.

1. Introduction

Bose–Einstein condensates (BECs) represent a unique state of matter formed at ultra-low temperatures, where bosons overlap and act collectively as a single quantum entity, providing a platform to explore phenomena such as superfluidity [1,2]. This phenomenon, predicted by Bose and Einstein in 1924–1925, was first demonstrated in 1995 using laser cooling techniques on various alkali atomic gases [3]. With optical trapping techniques replacing magnetic field trapping techniques, various spinor BECs have been observed in experiments, such as the spin-1 ^{23}Na condensate [4], the spin-1 ^{87}Rb condensate [5]. References for a comprehensive mathematical description of BECs and their corresponding analytical results can be found in [6–8]. Our main focus in this paper is to accurately compute the ground states of spin-1 BECs using numerical methods.

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For a spin-1 BEC, when the temperature is well below the critical temperature T_c, it can be described by a three-component vector wave function governed by the coupled Gross–Pitaevskii equations (GPEs) within the mean-field theory, as demonstrated in [9–11]. The ground state of a spin-1 BEC is defined as minimizing the associated energy functional of the coupled GPEs under two constraints: conservation of total mass and total magnetization. Within the framework of the Gross–Pitaevskii theory, numerous methods have been proposed over the past two decades to compute the ground states of spin-1 BECs. These methods can be classified into two categories: solving the energy minimization problem with constraints and solving the eigenvalue problem of the time-independent coupled GPEs. To solve the constrained energy minimization problem, the primary approach involves the normalized gradient flow (NGF) methods [12–16], along with other optimization techniques [17–21]. For the eigenvalue problem of the time-independent coupled GPEs, we refer readers to the literature [22–24].

Our primary focus is on the NGF methods, a class of widely used techniques for computing the ground states of spin-1 BECs. These include the gradient flow with discrete normalization (GFDN), the continuous normalized gradient flow (CNGF), and the normalized gradient flow with a Lagrange multiplier (GFLM). The central idea involves computing ground states by obtaining steady-state solutions to the partial differential equations derived from the variational energy functional of the GPEs, while enforcing normalization constraints through constraint-preserving techniques. The CNGF [12] can be viewed as introducing Lagrange multipliers into the gradient flow to enforce mass and magnetization conservation constraints. However, the CNGF requires a fully implicit discretization scheme to satisfy the constraint conditions, which leads to low computational efficiency. To address this limitation, the GFDN incorporates an additional projection step after gradient flow evolution to rigorously enforce mass and magnetization conservation constraints. Liu and Cai recently revisited the GFDN method for computing the ground states of spin-1 BECs [15], discovering that its temporal discretizations often yield incorrect ground states due to inherent artificial splitting errors comparable to the time step size. To accurately and efficiently compute the ground states of spin-1 BECs, they [15] propose the GFLM method, which can be viewed as an approximation of the CNGF by incorporating explicit Lagrange multiplier terms into the GFDN. In this manner, the steady-state solution of GFLM can be effectively computed through semi-implicit schemes.

Although the computation at each iteration of the GFLM method is efficient when an appropriate time discretization is chosen, its slow convergence speed requires many iterations to reach the ground state, resulting in excessive computational costs. Therefore, we aim to develop an enhanced computational method for the ground states of spin-1 BECs that retains GFLM's per-iteration efficiency while significantly reducing the number of iterations, thereby improving overall computational performance.

Chen et al. [25] propose an accelerated gradient flow method to compute the ground state of rotating BECs. Their approach essentially applies Nesterov's acceleration technique to an unconstrained Gross–Pitaevskii energy functional incorporating rotational terms, then enforces mass conservation via a Lagrange multiplier during the evolution. The accelerated gradient flow's rapid convergence enables their method to compute ground state solutions efficiently. The connection between accelerated gradient flow and Nesterov's accelerated method is explained by Su et al. in [26]. They derive a second-order ordinary differential equation (ODE) that serves as the exact limit of Nesterov's accelerated method. Subsequently, numerous studies have focused on inertial dynamics through second-order ODEs to understand Nesterov's accelerated method, as highlighted in [27–30] and related references.

Inspired by their work, we propose a continuous normalized accelerated gradient flow (CNAGF) to approximate the ground states of spin-1 BECs. This process can be interpreted as applying Nesterov's accelerated method to an unconstrained energy functional of coupled GPEs, followed by the introduction of Lagrange multipliers to ensure that the entire flow trajectory satisfies the constraints of mass and magnetization conservation. By solving the steady-state solution of the CNAGF under these constraints, we can compute the ground states of spin-1 BECs. Due to the integral terms included in the Lagrange multiplier terms of the CNAGF, the results obtained using explicit and semi-implicit schemes often fail to satisfy the constraints of mass and magnetization conservation, making it impossible to accurately compute the ground states of spin-1 BECs. Enforcing these constraints by adding an additional projection step to the CNAGF (similar to the GFLM method) is highly challenging, due to the lack of precisely determined projection coefficients. Therefore, we need a novel constraint-preserving approach to efficiently solve the CNAGF.

We find that coupling CNAGF with mass and magnetization conservation constraints and incorporating appropriate supplementary variables constructs a well-defined, solvable system, termed the reconfigured continuous normalized accelerated gradient flow (RCNAGF). When solving the steady-state solutions of RCNAGF to compute the ground states of spin-1 BECs, the results always satisfy the conservation of mass and magnetization constraints. The construction of RCNAGF is inspired by the supplementary variables method (SVM) introduced by Gong et al. [31,32]. To compare the computational efficiency with the GFLM method for solving the ground states of spin-1 BECs, we similarly employ explicit and semi-implicit schemes for temporal discretization and use spectral methods in the spatial direction to solve RCNAGF. During each time step, the explicit scheme requires a straightforward update, while the semi-implicit schemes involve solving linear systems with constant coefficients using a stabilization technique. The supplementary variables can be obtained by solving a low-dimensional algebraic system using Newton's iterative method. By selecting appropriate supplied functions, it is crucial to ensure that the final form of our schemes matches the corresponding Euler–Lagrange equations when evolution terminates. This ensures that our schemes can accurately compute the ground states of spin-1 BECs.

Our main contribution in this paper can be summarized as follows.

1. We propose a new generalized inertial dynamics model for simulating the ground states of spin-1 BECs, termed the CNAGF.

2. Building upon CNAGF, we develop the RCNAGF method with enforced mass and magnetization conservation to accurately compute ground states of spin-1 BECs.

3. Numerical experiments show that under identical time steps and convergence criteria, the RCNAGF-based semi-implicit scheme requires significantly fewer iterations than the GFLM method, while its explicit scheme allows larger time steps, leading to substantially improved computational efficiency compared to GFLM.

The remainder of this paper is structured as follows. In Section 2, we outline some preliminaries on the ground states of spin-1 BECs and accelerated gradient flows. In Section 3, we introduce the CNAGF and RCNAGF to simulate the ground states of spin-1 BECs. In Section 4, we propose explicit and semi-implicit schemes to solve the RCNAGF and prove that these schemes accurately compute ground states when appropriate supplementary functions are selected. In Section 5, we present numerical experiments validating the efficiency and accuracy of our methods. Concluding remarks are provided in Section 6.

2. Preliminaries

In this section, we primarily focus on reviewing fundamental knowledge regarding the ground states of spin-1 BECs and accelerated gradient flows. This review serves as necessary preparation for deriving the CNAGF and RCNAGF.

2.1. The ground states of spin-1 BECs

We consider the dimensionless coupled GPEs in d dimensions ($d = 1, 2, 3$)

$$i\boldsymbol{\Psi}_t = \left[-\frac{1}{2}\Delta + V(\mathbf{x}) + A(\boldsymbol{\Psi}) + B(\boldsymbol{\Psi})\right]\boldsymbol{\Psi}, \tag{2.1}$$

where $\boldsymbol{\Psi}(\mathbf{x},t) = (\psi_1(\mathbf{x},t), \psi_0(\mathbf{x},t), \psi_{-1}(\mathbf{x},t))^T \in (H^1(\Omega))^3$ are the three-component wave functions, Δ is the Laplace operator and $V(\mathbf{x}) \geq 0$ is a real-valued potential whose shape is determined by the type of system under investigation. The Hermitian matrices $A(\boldsymbol{\Psi})$ and $B(\boldsymbol{\Psi})$ are given below:

$$A(\boldsymbol{\Psi}) = \begin{pmatrix} p_1(\boldsymbol{\Psi}) & 0 & 0 \\ 0 & p_0(\boldsymbol{\Psi}) & 0 \\ 0 & 0 & p_{-1}(\boldsymbol{\Psi}) \end{pmatrix}, \qquad B(\boldsymbol{\Psi}) = \beta_n \begin{pmatrix} 0 & \psi_0\bar{\psi}_{-1} & 0 \\ \bar{\psi}_0\psi_{-1} & 0 & \psi_1\bar{\psi}_0 \\ 0 & \bar{\psi}_1\psi_0 & 0 \end{pmatrix}, \tag{2.2}$$

with

$$\begin{cases} p_1(\boldsymbol{\Psi}) = (\beta_n + \beta_s)\left(|\psi_1|^2 + |\psi_0|^2\right) + (\beta_n - \beta_s)|\psi_{-1}|^2, \\ p_0(\boldsymbol{\Psi}) = \left(\beta_n + \beta_s\right)\left(|\psi_1|^2 + |\psi_{-1}|^2\right) + \beta_n|\psi_0|^2, \\ p_{-1}(\boldsymbol{\Psi}) = (\beta_n + \beta_s)\left(|\psi_{-1}|^2 + |\psi_0|^2\right) + (\beta_n - \beta_s)|\psi_1|^2. \end{cases} \tag{2.3}$$

The parameters β_n and β_s refer to the dimensionless mean-field (spin-independent) and spin-exchange interaction, respectively, and $\bar{\psi}$ denotes the complex conjugate of the wave function ψ.

In this paper, we argue that Ω is bounded and its boundary condition is periodic. We use $(\cdot,\cdot)$ to denote the L^2 inner product and the L^2 norm $\|\cdot\|$ is reduced from this inner product, i.e.,

$$(u, v) = \int_\Omega \overline{u(\mathbf{x})}v(\mathbf{x})d\mathbf{x} \quad \text{and} \quad \|u\|^2 = (u,u), \quad \forall u, v \in L^2(\Omega).$$

The inner product and norm of vector function $\mathbf{u} = (u_1, u_0, u_{-1})^T$ and $\mathbf{v} = (v_1, v_0, v_{-1})^T$ are defined as:

$$(\mathbf{u},\mathbf{v}) = \int_\Omega \sum_{j=-1}^{1} \bar{u}_j v_j d\mathbf{x} \quad \text{and} \quad \|\mathbf{u}\|^2 = (\mathbf{u},\mathbf{u}).$$

Three important invariants of (2.1) are the mass (or normalization) of the wave function

$$N(\boldsymbol{\Psi}(\cdot,t)) := \|\boldsymbol{\Psi}\|^2 := \int_\Omega \sum_{j=-1}^{1} |\psi_j(\mathbf{x})|^2 d\mathbf{x} := \sum_{j=-1}^{1} \|\psi_j\|^2 = N(\boldsymbol{\Psi}(\cdot,0)) = 1, \tag{2.4}$$

the magnetization (with $m \in [-1, 1]$)

$$M(\boldsymbol{\Psi}(\cdot,t)) := \int_\Omega \left[|\psi_1(\mathbf{x})|^2 - |\psi_{-1}(\mathbf{x})|^2\right] d\mathbf{x} := \|\psi_1\|^2 - \|\psi_{-1}\|^2 = M(\boldsymbol{\Psi}(\cdot,0)) = m, \tag{2.5}$$

and the energy per particle

$$\begin{aligned} E(\boldsymbol{\Psi}(\cdot,t)) = &\int_\Omega \sum_{l=-1}^{1}\left(\frac{1}{2}|\nabla\psi_l|^2 + V(\mathbf{x})|\psi_l|^2\right) + (\beta_n - \beta_s)|\psi_1|^2|\psi_{-1}|^2 + \frac{\beta_n}{2}|\psi_0|^4 \\ &+ \frac{\beta_n + \beta_s}{2}\left[|\psi_1|^4 + |\psi_{-1}|^4 + 2|\psi_0|^2\left(|\psi_1|^2 + |\psi_{-1}|^2\right)\right] + \beta_s\left(\bar{\psi}_{-1}\psi_0^2\bar{\psi}_1 + \psi_{-1}\bar{\psi}_0^2\psi_1\right)d\mathbf{x} \\ = &\,E(\boldsymbol{\Psi}(\cdot,0)). \end{aligned} \tag{2.6}$$

Mathematically, the ground state of a spin-1 BEC can be modeled as a minimizer of the energy functional (2.6) of the coupled GPEs under normalization and magnetization constraints. It can be defined as the minimizer of the following nonconvex minimization problem:

Find ($\boldsymbol{\Phi}_g = (\phi_1^g(\mathbf{x}), \phi_0^g(\mathbf{x}), \phi_{-1}^g(\mathbf{x}))^T \in G$) such that

$$E(\boldsymbol{\Phi}_g) := \min_{\boldsymbol{\Phi}_g \in G} E(\boldsymbol{\Phi}_g) \tag{2.7}$$

where the energy function $E(\mathbf{\Phi})$ is defined as (2.6) and the nonconvex set is

$$G = \left\{ \mathbf{\Phi} = (\phi_1, \phi_0, \phi_{-1})^T \in (H^1(\Omega))^3 \middle| \|\mathbf{\Phi}\|^2 = 1, \|\phi_1\|^2 - \|\phi_{-1}\|^2 = m, E(\mathbf{\Phi}) < \infty \right\}.$$

The standard theory [33] guarantees the existence of a minimizer for above nonconvex minimization problem when $\beta_n \geq 0$, $\beta_n \geq |\beta_s|$, and $\lim_{|\mathbf{x}|\to\infty} V(\mathbf{x}) = \infty$. For more information on the analytical properties of spin-1 BECs' ground states, we refer to [6,34–36].

As stated in [13], the definition of Lagrangian is

$$\mathcal{L}(\mathbf{\Phi}, \mu, \lambda) := E(\mathbf{\Phi}) - \mu \left(\|\phi_1\|^2 + \|\phi_0\|^2 + \|\phi_{-1}\|^2 - 1 \right) - \lambda \left(\|\phi_1\|^2 - \|\phi_{-1}\|^2 - m \right). \tag{2.8}$$

Differentiating (2.8) with respect to $\bar{\phi}_1$, $\bar{\phi}_0$, and $\bar{\phi}_{-1}$, respectively, the associated Euler–Lagrange equations can be established as follows:

$$(\mu I_3 + \lambda C)\mathbf{\Phi} = \left[-\frac{1}{2}\Delta + V(\mathbf{x}) + A(\mathbf{\Phi}) + B(\mathbf{\Phi}) \right] \mathbf{\Phi} := H(\mathbf{\Phi})\mathbf{\Phi}, \tag{2.9}$$

where μ and λ are the Lagrange multipliers (or chemical potentials) of the coupled GPEs (2.9) that are related to the normalization constraint (2.4) and the magnetization constraint (2.5). The I_3 is a 3-by-3 identity matrix and C is given by $\mathrm{diag}(1, 0, -1)$.

It is clear that (2.9) is also a nonlinear eigenvalue problem under the following two constraints

$$\|\mathbf{\Phi}\|^2 = 1, \quad \text{and} \quad \|\phi_1\|^2 - \|\phi_{-1}\|^2 = m. \tag{2.10}$$

The nonlinear eigenvalue problem (2.9) can be obtained by substituting $\psi_j(\mathbf{x}, t) = e^{-i\mu_j t}\phi_j(\mathbf{x})$ $(j = 1, 0, -1)$ with $\mu_1 = \mu + \lambda$, $\mu_0 = \mu$ and $\mu_{-1} = \mu - \lambda$ in the coupled GPEs (2.1). Clearly, these three eigenvalues satisfy relationship $\mu_1 + \mu_{-1} = 2\mu_0$.

Now, we review the CNGF, proposed by Bao and Wang in [13], for computing the ground states of spin-1 BECs. Its specific form is as follows:

$$\partial_t \mathbf{\Phi}(\mathbf{x}, t) = \left[\frac{1}{2}\Delta - V(\mathbf{x}) - A(\mathbf{\Phi}) - B(\mathbf{\Phi}) + \mu_{\mathbf{\Phi}}(t) I_3 + \lambda_{\mathbf{\Phi}}(t) C \right] \mathbf{\Phi}(\mathbf{x}, t), \tag{2.11}$$

where the Lagrange multipliers $\mu_{\mathbf{\Phi}}(t)$ and $\lambda_{\mathbf{\Phi}}(t)$ are introduced to ensure that the flow preserves the constraints of mass and magnetization, respectively. They are given as

$$\mu_{\mathbf{\Phi}}(t) = \frac{R_{\mathbf{\Phi}}(t) D_{\mathbf{\Phi}}(t) - M_{\mathbf{\Phi}}(t) F_{\mathbf{\Phi}}(t)}{N_{\mathbf{\Phi}}(t) R_{\mathbf{\Phi}}(t) - M^2_{\mathbf{\Phi}}(t)} \quad \text{and} \quad \lambda_{\mathbf{\Phi}}(t) = \frac{N_{\mathbf{\Phi}}(t) F_{\mathbf{\Phi}}(t) - M_{\mathbf{\Phi}}(t) D_{\mathbf{\Phi}}(t)}{N_{\mathbf{\Phi}}(t) R_{\mathbf{\Phi}}(t) - M^2_{\mathbf{\Phi}}(t)}, \tag{2.12}$$

where

$$\begin{cases} M_{\mathbf{\Phi}}(t) = \|\phi_1\|^2 - \|\phi_{-1}\|^2, \quad R_{\mathbf{\Phi}}(t) = \|\phi_1\|^2 + \|\phi_{-1}\|^2, \\ N_{\mathbf{\Phi}}(t) = \|\phi_1\|^2 + \|\phi_0\|^2 + \|\phi_{-1}\|^2, \\ D_{\mathbf{\Phi}}(t) = \int_\Omega \mathbf{\Phi}^* H(\mathbf{\Phi}) \mathbf{\Phi} d\mathbf{x} \quad F_{\mathbf{\Phi}}(t) = \int_\Omega \Phi^* C H(\mathbf{\Phi}) \mathbf{\Phi} d\mathbf{x}. \end{cases} \tag{2.13}$$

Note that $\Phi^* = \bar{\Phi}^T$ is the conjugate transport.

The CNGF (2.11) can be viewed as applying the steepest descent method to the energy functional (2.6), and then enforcing the constraints of mass and magnetization conservation throughout the entire trajectory of the gradient flow by embedding Lagrange multipliers into the flow. It can be proved that the CNGF (2.11) is characterized by mass conservation (i.e., $N_{\mathbf{\Phi}}(t) = N_{\mathbf{\Phi}}(0) = 1$), magnetization conservation (i.e., $M_{\mathbf{\Phi}}(t) = M_{\mathbf{\Phi}}(0) = m$), and energy dissipation (i.e., $\frac{d}{dt}E(\mathbf{\Phi}) = -2\|\partial_t \mathbf{\Phi}\|^2 = -2(\|\partial_t \phi_1\|^2 + \|\partial_t \phi_0\|^2 + \|\partial_t \phi_{-1}\|^2)$) [13]. The CNGF (2.11) can be discretized via the Crank–Nicolson finite difference method with specialized treatment of the nonlinear terms [13]. This discretization yields an algorithm that preserves normalization and magnetization while exhibiting energy decay. However, the scheme is fully implicit, requiring the solution of a nonlinear system at each time step.

2.2. Accelerated gradient flow

The connection between ODE and numerical optimization is commonly made by using small step sizes, which ensures that the trajectory or solution path approximates a curve described by an ODE.

Consider the optimization problem

$$\min_{u \in R^n} f(u) \tag{2.14}$$

where $f : \mathbb{R}^n \longrightarrow \mathbb{R}$ is a Fréchet differentiable function with L-Lipschitz continuous gradient, i.e., for all $u, v \in \mathbb{R}^n$, exists $L \geq 0$ such that $\|\nabla f(u) - \nabla f(v)\| \leq L\|u - v\|$. Using the gradient descent method for minimizing objective function f, which reads

$$u_k = u_{k-1} - s\nabla f\left(u_{k-1}\right), \quad k = 1, 2, \ldots, \tag{2.15}$$

with $s \geq 0$ is a step length and the initial guess is $u_0 \in \mathbb{R}^n$. When the limit $s \to 0$, continuous-time limit of the gradient descent method is

$$u_t = -\nabla f(u(t)). \tag{2.16}$$

In [26], Su et al. propose the following dynamical system, known as the accelerated gradient flow, as the continuous-time limit of Nesterov's accelerated method:

$$\begin{cases} u_{tt} + \frac{\eta}{t}u_t + \nabla f(u(t)) = 0, \\ u(t_0) = u_0, \quad u_t(t_0) = 0, \end{cases} \tag{2.17}$$

where $t_0 \geq 0$ and $\eta \geq 3$. The corresponding Nesterov's acceleration method takes the specific form of

$$\begin{cases} u_k = w_{k-1} - s\nabla f\left(w_{k-1}\right), \\ w_k = u_k + \frac{k-1}{k+2}\left(u_k - u_{k-1}\right). \end{cases} \tag{2.18}$$

with the step size $s \geq 0$ and $k = 1, 2, \ldots$. For more detailed information on accelerated gradient flow, interested readers are referred to papers [28,37] and their references.

3. Accelerated gradient flows for computing spin-1 BEC ground states

In this section, we mainly propose CNAGF and RCNAGF to approach the ground states of spin-1 BECs.

3.1. Continuous normalized accelerated gradient flow

The CNAGF for simulating the ground states of spin-1 BECs can be viewed as applying the Nesterov's accelerated gradient method to the energy functional (2.6), where the mass and magnetization conservation constraints are enforced through Lagrange multipliers embedded along the entire flow trajectory. The specific form of CNAGF is as follows:

$$\begin{cases} \partial_{tt}\mathbf{\Phi}(\mathbf{x},t) + \frac{\eta}{t}\partial_t\mathbf{\Phi}(\mathbf{x},t) = \left[-H(\mathbf{\Phi}) + \xi_{\mathbf{\Phi}}(t)I_3 + \sigma_{\mathbf{\Phi}}(t)C\right]\mathbf{\Phi}(\mathbf{x},t), \quad \mathbf{x}\in\Omega, \quad t>0, \quad (a) \\ \mathbf{\Phi}(\mathbf{x},0) = \mathbf{\Phi}^0(\mathbf{x}) = (\phi_1^0(\mathbf{x}), \phi_0^0(\mathbf{x}), \phi_{-1}^0(\mathbf{x}))^T, \quad \partial_t\mathbf{\Phi}(\mathbf{x},0) = \mathbf{v}_0(\mathbf{x}) = \mathbf{0}, \quad \mathbf{x}\in\Omega, \quad (b) \\ N_{\mathbf{\Phi}}(t=0) := N_{\mathbf{\Phi}^0} := \|\phi_1^0\| + \|\phi_0^0\|^2 + \|\phi_{-1}^0\|^2 = 1, \qquad (c) \\ M_{\mathbf{\Phi}}(t=0) := M_{\mathbf{\Phi}^0} := \|\phi_1^0\| - \|\phi_{-1}^0\|^2 = m, \qquad (d) \end{cases} \tag{3.1}$$

where $\mathbf{\Phi}(\mathbf{x},t) = (\phi_1(\mathbf{x},t), \phi_0(\mathbf{x},t), \phi_{-1}(\mathbf{x},t))^T$ denotes the three-component wave function, with $\eta \geq 3$ being the damping coefficient and t is the artificial time variable. The operator $H(\mathbf{\Phi})\mathbf{\Phi}(\mathbf{x},t)$ is defined as (2.9). Here, $\mathbf{\Phi}^0(\mathbf{x})$ represents the initial state, $\mathbf{v}_0$ is the initial velocity, and the Lagrange multipliers $\xi_{\mathbf{\Phi}}(t)$ and $\sigma_{\mathbf{\Phi}}(t)$ ensure mass (normalization) and magnetization conservation in the CNAGF 3.1.

The specific expressions for the Lagrange multipliers $\xi_{\mathbf{\Phi}}(t)$ and $\sigma_{\mathbf{\Phi}}(t)$ can be derived through the following steps.

(1) Assume that the mass ($N_{\mathbf{\Phi}}(t) = \|\phi_1\|^2 + \|\phi_0\|^2 + \|\phi_{-1}\|^2$) of system 3.1 is conservation, we have

$$\begin{aligned} \frac{d}{dt}\left(N_{\mathbf{\Phi}}(t)\right) &= \int_\Omega \sum_{j=-1}^{1}\left[\overline{\phi_j}\partial_t\phi_j + \phi_j\partial_t\overline{\phi_j}\right]d\mathbf{x} := \left(\mathbf{\Phi}, \partial_t\mathbf{\Phi}\right) + \left(\partial_t\mathbf{\Phi}, \mathbf{\Phi}\right) = 0, \\ \frac{d^2}{dt^2}\left(N_{\mathbf{\Phi}}(t)\right) &= \int_\Omega \sum_{j=-1}^{1}\left[\overline{\phi_j}\partial_{tt}\phi_j + \phi_j\partial_{tt}\overline{\phi_j} + 2\partial_t\phi_j\partial_t\overline{\phi_j}\right]d\mathbf{x} \\ &:= \left(\mathbf{\Phi}, \partial_{tt}\mathbf{\Phi}\right) + \left(\partial_{tt}\mathbf{\Phi}, \mathbf{\Phi}\right) + 2\left\|\partial_t\mathbf{\Phi}\right\|^2 = 0. \end{aligned} \tag{3.2}$$

Similarly, we can obtain two additional conditions by assuming the conservation of magnetization ($M_{\mathbf{\Phi}}(t) = \|\phi_1\|^2 - \|\phi_{-1}\|^2$) in system 3.1 .

$$\begin{aligned} \frac{d}{dt}\left(M_{\mathbf{\Phi}}(t)\right) &= \int_\Omega \left[\overline{\phi_1}\partial_t\phi_1 + \phi_1\partial_t\overline{\phi_1}\right] - \left[\overline{\phi_{-1}}\partial_t\phi_{-1} + \phi_{-1}\partial_t\overline{\phi_{-1}}\right]d\mathbf{x} = \left(C\mathbf{\Phi}, \partial_t\mathbf{\Phi}\right) + \left(\partial_t\mathbf{\Phi}, C\mathbf{\Phi}\right) = 0, \\ \frac{d^2}{dt^2}\left(M_{\mathbf{\Phi}}(t)\right) &= \int_\Omega \left[\overline{\phi_1}\partial_{tt}\phi_1 + \phi_1\partial_{tt}\overline{\phi_1} + 2\partial_t\phi_1\partial_t\overline{\phi_1}\right] - \left[\overline{\phi_{-1}}\partial_{tt}\phi_{-1} + \phi_{-1}\partial_{tt}\overline{\phi_{-1}} + 2\partial_t\phi_{-1}\partial_t\overline{\phi_{-1}}\right]d\mathbf{x} \\ &:= \left(C\mathbf{\Phi}, \partial_{tt}\mathbf{\Phi}\right) + \left(\partial_{tt}\mathbf{\Phi}, C\mathbf{\Phi}\right) + 2\left(\partial_t\mathbf{\Phi}, C\partial_t\mathbf{\Phi}\right) = 0. \end{aligned} \tag{3.3}$$

(2) Taking the L^2 inner products with $\mathbf{\Phi}$ on both sides of (3.1(a)) and then conjugate of the result, yield

$$\left(\mathbf{\Phi}, \partial_{tt}\mathbf{\Phi}\right) + \frac{\eta}{t}\left(\mathbf{\Phi}, \partial_t\mathbf{\Phi}\right) = \left(\mathbf{\Phi}, \left[\frac{1}{2}\Delta - V(\mathbf{x}) - A(\mathbf{\Phi}) - B(\mathbf{\Phi}) + \xi_{\mathbf{\Phi}}(t)I_3 + \sigma_{\mathbf{\Phi}}(t)C\right]\mathbf{\Phi}\right), \tag{3.4}$$

and

$$\left(\partial_{tt}\mathbf{\Phi}, \mathbf{\Phi}\right) + \frac{\eta}{t}\left(\partial_t\mathbf{\Phi}, \mathbf{\Phi}\right) = \left(\left[\frac{1}{2}\Delta - V(\mathbf{x}) - A(\mathbf{\Phi}) - B(\mathbf{\Phi}) + \xi_{\mathbf{\Phi}}(t)I_3 + \sigma_{\mathbf{\Phi}}(t)C\right]\mathbf{\Phi}, \mathbf{\Phi}\right). \tag{3.5}$$

(3) Substituting (3.2) into (3.4) and (3.5), we derive

$$\begin{aligned} 0 &= \left(\mathbf{\Phi}, \partial_{tt}\mathbf{\Phi}\right) + \left(\partial_{tt}\mathbf{\Phi}, \mathbf{\Phi}\right) + \frac{\eta}{t}\left[\left(\mathbf{\Phi}, \partial_t\mathbf{\Phi}\right) + \left(\partial_t\mathbf{\Phi}, \mathbf{\Phi}\right)\right] + 2\left\|\partial_t\mathbf{\Phi}\right\|^2 \\ &= \left(\mathbf{\Phi}, \left[\frac{1}{2}\Delta - V(\mathbf{x}) - A(\mathbf{\Phi}) - B(\mathbf{\Phi}) + \xi_{\mathbf{\Phi}}(t)I_3 + \sigma_{\mathbf{\Phi}}(t)C\right]\mathbf{\Phi}\right) \\ &\quad + \left(\left[\frac{1}{2}\Delta - V(\mathbf{x}) - A(\mathbf{\Phi}) - B(\mathbf{\Phi}) + \xi_{\mathbf{\Phi}}(t)I_3 + \sigma_{\mathbf{\Phi}}(t)C\right]\mathbf{\Phi}, \mathbf{\Phi}\right) + 2\left\|\partial_t\mathbf{\Phi}\right\|^2 \\ &= -2D_{\mathbf{\Phi}}(t) + 2\xi_{\mathbf{\Phi}}(t)N_{\mathbf{\Phi}}(t) + 2\sigma_{\mathbf{\Phi}}(t)M_{\mathbf{\Phi}}(t) + 2\left\|\partial_t\mathbf{\Phi}\right\|^2, \end{aligned} \tag{3.6}$$

where $D_{\mathbf{\Phi}}(t) = \int_\Omega \mathbf{\Phi}^* H(\mathbf{\Phi})\mathbf{\Phi} d\mathbf{x} = \int_\Omega (H(\mathbf{\Phi})\mathbf{\Phi})^*\mathbf{\Phi} d\mathbf{x}$.

(4) Substituting (3.3) into (3.4) and (3.5), we derive

$$\begin{aligned} 0 &= \left(C\mathbf{\Phi}, \partial_{tt}\mathbf{\Phi}\right) + \left(\partial_{tt}\mathbf{\Phi}, C\mathbf{\Phi}\right) + \frac{\eta}{t}\left[\left(C\mathbf{\Phi}, \partial_t\mathbf{\Phi}\right) + \left(\partial_t\mathbf{\Phi}, C\mathbf{\Phi}\right)\right] + 2\left(C\partial_t\mathbf{\Phi}, \partial_t\mathbf{\Phi}\right) \\ &= \left(C\mathbf{\Phi}, \left[\frac{1}{2}\Delta - V(\mathbf{x}) - A(\mathbf{\Phi}) - B(\mathbf{\Phi}) + \xi_{\mathbf{\Phi}}(t)I_3 + \sigma_{\mathbf{\Phi}}(t)C\right]\mathbf{\Phi}\right) \\ &\quad + \left(\left[\frac{1}{2}\Delta - V(\mathbf{x}) - A(\mathbf{\Phi}) - B(\mathbf{\Phi}) + \xi_{\mathbf{\Phi}}(t)I_3 + \sigma_{\mathbf{\Phi}}(t)C\right]\mathbf{\Phi}, C\mathbf{\Phi}\right) + 2\left(C\partial_t\mathbf{\Phi}, \partial_t\mathbf{\Phi}\right) \\ &= -2F_{\mathbf{\Phi}}(t) + 2\xi_{\mathbf{\Phi}}(t)M_{\mathbf{\Phi}}(t) + 2\sigma_{\mathbf{\Phi}}(t)R_{\mathbf{\Phi}}(t) + 2\left(C\partial_t\mathbf{\Phi}, \partial_t\mathbf{\Phi}\right), \end{aligned} \tag{3.7}$$

where $R_{\mathbf{\Phi}}(t) = \|\phi_1\|^2 + \|\phi_{-1}\|^2$ and $F_{\mathbf{\Phi}}(t) = \int_\Omega \mathbf{\Phi}^* C H(\mathbf{\Phi})\mathbf{\Phi} d\mathbf{x} = \int_\Omega \left(C H(\mathbf{\Phi})\mathbf{\Phi}\right)^*\mathbf{\Phi} d\mathbf{x}$.

(5) Combining (3.6) and (3.7), we get

$$\begin{cases} -D_{\mathbf{\Phi}}(t) + \xi_{\mathbf{\Phi}}(t)N_{\mathbf{\Phi}}(t) + \sigma_{\mathbf{\Phi}}(t)M_{\mathbf{\Phi}}(t) + (\partial_t\mathbf{\Phi}, \partial_t\mathbf{\Phi}) = 0 \\ -F_{\mathbf{\Phi}}(t) + \xi_{\mathbf{\Phi}}(t)M_{\mathbf{\Phi}}(t) + \sigma_{\mathbf{\Phi}}(t)R_{\mathbf{\Phi}}(t) + (C\partial_t\mathbf{\Phi}, \partial_t\mathbf{\Phi}) = 0 \end{cases}$$

By computing, we drive

$$\xi_{\mathbf{\Phi}}(t) = \frac{R_{\mathbf{\Phi}}(t)[D_{\mathbf{\Phi}}(t) - (\partial_t\mathbf{\Phi}, \partial_t\mathbf{\Phi})] - M_{\mathbf{\Phi}}(t)[F_{\mathbf{\Phi}}(t) - (C\partial_t\mathbf{\Phi}, \partial_t\mathbf{\Phi})]}{N_{\mathbf{\Phi}}(t)R_{\mathbf{\Phi}}(t) - M_{\mathbf{\Phi}}^2(t)} \tag{3.8}$$

and

$$\sigma_{\mathbf{\Phi}}(t) = \frac{N_{\mathbf{\Phi}}(t)[F_{\mathbf{\Phi}}(t) - (C\partial_t\mathbf{\Phi}, \partial_t\mathbf{\Phi})] - M_{\mathbf{\Phi}}(t)[D_{\mathbf{\Phi}}(t) - (\partial_t\mathbf{\Phi}, \partial_t\mathbf{\Phi})]}{N_{\mathbf{\Phi}}(t)R_{\mathbf{\Phi}}(t) - M_{\mathbf{\Phi}}^2(t)}. \tag{3.9}$$

Formally, at the steady state of the flow 3.1, all the time derivative terms vanish, i.e. $\partial_t\mathbf{\Phi} = (\partial_t\phi_1, \partial_t\phi_0, \partial_t\phi_{-1})^T = \mathbf{0}$ and $\partial_{tt}\mathbf{\Phi} = (\partial_{tt}\phi_1, \partial_{tt}\phi_0, \partial_{tt}\phi_{-1})^T = \mathbf{0}$, which leads to $\xi_{\mathbf{\Phi}}(t) = \mu_{\mathbf{\Phi}}(t)$ and $\sigma_{\mathbf{\Phi}}(t) = \lambda_{\mathbf{\Phi}}(t)$. This means that the evolution of CNAGF 3.1 is the same as the CNGF (2.11), which can accurately calculate the spin-1 BECs ground states. In other words, the steady state equation of (3.1(*a*)) is exactly the Euler–Lagrange Eq. (2.9)

$$\left(\xi_{\mathbf{\Phi}}(t)I_3 + \sigma_{\mathbf{\Phi}}(t)C\right)\mathbf{\Phi} = \left(\mu_{\mathbf{\Phi}}(t)I_3 + \lambda_{\mathbf{\Phi}}(t)C\right)\mathbf{\Phi} = \left[-\frac{1}{2}\Delta + V(\mathbf{x}) + A(\mathbf{\Phi}) + B(\mathbf{\Phi})\right]\mathbf{\Phi}.$$

We define the total cost function linked to the solution $\mathbf{\Phi} = \mathbf{\Phi}(\mathbf{x}, t)$ of CANGF 3.1 as

$$L(\mathbf{\Phi}(\cdot, t)) = E(\mathbf{\Phi}(\cdot, t)) + \|\partial_t\mathbf{\Phi}(\cdot, t)\|^2. \tag{3.10}$$

The subsequent theorem demonstrates that, under specific conditions, the CNAGF 3.1 is inherently conservative with respect to mass and magnetization, and exhibits a dissipative property with respect to $L(\mathbf{\Phi}(\cdot, t))$.

Theorem 3.1. *Let $\mathbf{\Phi}(\mathbf{x}, t) = (\phi_1(\mathbf{x}, t), \phi_0(\mathbf{x}, t), \phi_{-1}(\mathbf{x}, t))^T$ be the solution to the CNAGF 3.1 with the Lagrange multipliers $\xi_{\mathbf{\Phi}}(t)$ and $\sigma_{\mathbf{\Phi}}(t)$ determined by (3.8) and (3.9), respectively. If the each component of $\mathbf{\Phi}(\mathbf{x}, t)$ is $\phi_j \in \mathbb{C}^2((0, \infty), X)$ $(j = -1, 0, 1)$, where $X = \{\phi_j \in H^1(\Omega) : E(\mathbf{\Phi}) < \infty\}$, then for any given initial data $\mathbf{\Phi}(\mathbf{x}, 0) = \mathbf{\Phi}^0(\mathbf{x})$ satisfying $N_{\mathbf{\Phi}^0} = 1$, $M_{\mathbf{\Phi}^0} = m$ and $\partial_t\mathbf{\Phi}^0 := v_{\mathbf{0}}(\mathbf{x}) = \mathbf{0}$, we have*

$$\begin{cases} N_{\mathbf{\Phi}^0} = N_{\mathbf{\Phi}}(t) = 1, \quad t \geq 0 \\ M_{\mathbf{\Phi}^0} = M_{\mathbf{\Phi}}(t) = m, \quad t \geq 0 \\ L(\mathbf{\Phi}(\cdot, t)) \leq L(\mathbf{\Phi}(\cdot, s)), \quad \forall t \geq s \geq 0. \end{cases} \tag{3.11}$$

Proof. Substituting $\xi_{\mathbf{\Phi}}(t)$(3.8) and $\sigma_{\mathbf{\Phi}}(t)$(3.9) into the sum of (3.4) and (3.5), we obtain

$$\left(\mathbf{\Phi}, \partial_{tt}\mathbf{\Phi}\right) + \frac{\eta}{t}\left(\mathbf{\Phi}, \partial_t\mathbf{\Phi}\right) + \left(\partial_{tt}\mathbf{\Phi}, \mathbf{\Phi}\right) + \frac{\eta}{t}\left(\partial_t\mathbf{\Phi}, \mathbf{\Phi}\right) = -2\|\partial_t\mathbf{\Phi}\|^2.$$

After rearranging, we obtain

$$\begin{aligned} \frac{d}{dt}(\partial_t\|\mathbf{\Phi}\|^2) &= \frac{d}{dt}\left[\left(\mathbf{\Phi}, \partial_t\mathbf{\Phi}\right) + \left(\partial_t\mathbf{\Phi}, \mathbf{\Phi}\right)\right] = \left(\mathbf{\Phi}, \partial_{tt}\mathbf{\Phi}\right) + \left(\partial_{tt}\mathbf{\Phi}, \mathbf{\Phi}\right) + 2\|\partial_t\mathbf{\Phi}\|^2 \\ &= -\frac{\eta}{t}\left(\mathbf{\Phi}, \partial_t\mathbf{\Phi}\right) - \frac{\eta}{t}\left(\partial_t\mathbf{\Phi}, \mathbf{\Phi}\right) = -\frac{\eta}{t}(\partial_t\|\mathbf{\Phi}\|^2). \end{aligned}$$

Since $\mathbf{v}_0(\mathbf{x}) = \mathbf{0}$ and $\frac{\eta}{t} > 0$ is locally integrable on $(0, t]$, we have

$$\partial_t \|\mathbf{\Phi}(\cdot, t)\|^2 = \lim_{\varepsilon \to 0^+} e^{-\int_\varepsilon^t \frac{\eta}{s} ds} \partial_t \|\mathbf{\Phi}(\cdot, \varepsilon)\|^2 = 0, \quad \forall t > 0.$$

Therefore, we have $\frac{d}{dt}\left(N_{\mathbf{\Phi}}(t)\right) = 0$, which confirms the conservation of mass. The proof of magnetization conservation is similar, and we omit the proof process here. Additionally, by employing (3.1(a)) and recognizing that $\partial_t \|\boldsymbol{\Phi}(\cdot, \mathbf{t})\|^2 = 0$, a straightforward computation demonstrates that

$$\begin{aligned}
&\frac{d}{dt} L(\mathbf{\Phi}(\cdot, t)) = \frac{d}{dt}\Big(E(\mathbf{\Phi}(\cdot, t)) + \|\partial_t \mathbf{\Phi}(\cdot, t)\|^2\Big) \\
&= \Big(\partial_t \mathbf{\Phi}, \partial_{tt} \mathbf{\Phi}\Big) - \Big(\partial_t \mathbf{\Phi}, \Big[\frac{1}{2}\Delta - V(\mathbf{x}) - A(\mathbf{\Phi}) - B(\mathbf{\Phi})\Big] \mathbf{\Phi}\Big) \\
&\quad + \Big(\partial_{tt} \mathbf{\Phi}, \partial_t \mathbf{\Phi}\Big) - \Big(\Big[\frac{1}{2}\Delta - V(\mathbf{x}) - A(\mathbf{\Phi}) - B(\mathbf{\Phi})\Big] \mathbf{\Phi}, \partial_t \mathbf{\Phi}\Big) \\
&= \Big(\partial_t \mathbf{\Phi}, -\frac{\eta}{t}\partial_t \mathbf{\Phi} + \xi_{\mathbf{\Phi}}(t) I_3 \boldsymbol{\Phi} + \sigma_{\mathbf{\Phi}}(t) C \mathbf{\Phi}\Big) + \Big(-\frac{\eta}{t}\partial_t \mathbf{\Phi} + \xi_{\mathbf{\Phi}}(t) I_3 \mathbf{\Phi} + \sigma_{\mathbf{\Phi}}(t) C \mathbf{\Phi}, \partial_t \mathbf{\Phi}\Big) \\
&= -\frac{2\eta}{t}\|\partial_t \mathbf{\Phi}\|^2 + (\xi_{\mathbf{\Phi}}(t) I_3 + \sigma_{\mathbf{\Phi}}(t) C)\Big[\Big(\partial_t \mathbf{\Phi}, \mathbf{\Phi}\Big) + \Big(\mathbf{\Phi}, \partial_t \mathbf{\Phi}\Big)\Big] \\
&= -\frac{2\eta}{t}\|\partial_t \mathbf{\Phi}\|^2 \le 0.
\end{aligned}$$

This implies the dissipation of $L(\mathbf{\Phi}(\cdot, t))$, thus concluding the proof.

Specifically, with the initial velocity taken as $\mathbf{v}_0(\mathbf{x}) = \mathbf{0}$, the entire flow trajectory 3.1 is naturally contained in the energy sublevel set $\{\mathbf{\Phi} \in G : E(\mathbf{\Phi}) \le E(\mathbf{\Phi}_0)\}$. This provides some local stability to the CNAGF 3.1 for computing the ground states of spin-1 BECs.

3.2. Reconfigured continuous normalized accelerated gradient flow

To efficiently compute the ground state of spin-1 BECs, we employ explicit and semi-implicit schemes to circumvent the need for iterative solution of nonlinear systems at each time step. The presence of integral terms in the Lagrange multipliers $\xi_{\mathbf{\Phi}}(t)$ and $\sigma_{\mathbf{\Phi}}(t)$ poses a significant challenge for constructing mass- and magnetization-preserving numerical schemes when solving the CNAGF. Therefore, we adapt the core concept of the supplementary variables method (SVM), which is a structured algorithm originally developed to maintain the energy dissipation rate of dynamical systems, to construct a well-defined system that rigorously enforces mass and magnetization conservation constraints.

The SVM introduces supplementary variables (equal in number to the deduced equations) into the over-determined system of differential equations. This reformulation transforms the system into a structurally stable and well-posed system within a higher-dimensional phase space. Specifically, it involves embedding the original system into a higher-dimensional reformulated system where the solutions lie within a stable manifold. Here, we construct the RCANGF by coupling CNAGF 3.1 with mass and magnetization conservation constraints and incorporating appropriate supplementary variables to simulate the ground state of spin-1 BECs.

$$\begin{cases}
\partial_{tt} \mathbf{\Phi}(\mathbf{x}, t) + \frac{\eta}{t}\partial_t \mathbf{\Phi}(\mathbf{x}, t) = \left[-H(\mathbf{\Phi}) + \xi_{\mathbf{\Phi}}(t) I_3 + \sigma_{\mathbf{\Phi}}(t) C\right] \mathbf{\Phi}(\mathbf{x}, t) + \alpha \mathbf{\Xi}(\mathbf{\Phi}) + \gamma \mathbf{\Upsilon}(\mathbf{\Phi}), \\
N_{\mathbf{\Phi}}(t) := \|\phi_1\|^2 + \|\phi_0\|^2 + \|\phi_{-1}\|^2 = 1, \\
M_{\mathbf{\Phi}}(t) := \|\phi_1\|^2 - \|\phi_{-1}\|^2 = m, \\
\mathbf{\Phi}(\mathbf{x}, 0) = \mathbf{\Phi}^0(\mathbf{x}), \quad \partial_t \mathbf{\Phi}(\mathbf{x}, 0) = \mathbf{v}^0(\mathbf{x}) = \mathbf{0}, \quad \mathbf{x} \in \Omega, \\
M_{\boldsymbol{\Phi}^0} = m \quad \text{and} \quad N_{\boldsymbol{\Phi}^0} = 1,
\end{cases} \tag{3.12}$$

where $\eta \ge 3$ is a damping coefficient and t denotes the artificial time variable. The operator $H(\mathbf{\Phi})\mathbf{\Phi}(\mathbf{x}, t)$ is defined as (2.9), while the Lagrange multipliers $\xi_{\mathbf{\Phi}}(t)$ and $\sigma_{\mathbf{\Phi}}(t)$ are given by (3.8) and (3.9), respectively. The parameters α and γ represent supplementary variables.

$$\mathbf{\Xi}(\mathbf{\Phi}) = g_1(\phi_1), g_0(\phi_0), g_{-1}(\phi_{-1})^T \quad \text{and} \quad \mathbf{\Upsilon}(\mathbf{\Phi}) = g_2(\phi_1), 0, g_{-2}(\phi_{-1})^T$$

are column vectors representing the complementary functions of a storage system, where each element corresponds to the complementary function of each component.

After discretizing RCNAGF (3.12) in the temporal direction, we obtain a stable and solvable system consisting of five unknowns and five equations. In theory, we are searching for the solution $\mathbf{\Phi}$ of the RCNAGF (3.12) at $\alpha = \gamma = 0$. At the system (3.12) stops evolution, the function $\mathbf{\Phi}_g$ must minimize the energy $E(\mathbf{\Phi})$ on the manifold G, which implies that $\partial_t \mathbf{\Phi}|_{\mathbf{\Phi}=\mathbf{\Phi}_g} = \alpha \mathbf{\Xi}(\mathbf{\Phi}_g) + \gamma \mathbf{\Upsilon}(\mathbf{\Phi}_g) = \mathbf{0}$. This means that the supplementary variables α, γ must be 0. That is, we design our schemes with a stopping criterion of $\|\boldsymbol{\Phi}^{n+1} - \boldsymbol{\Phi}^n\|_\infty < \epsilon$, controlling the precision through the parameter ϵ, and the supplementary variables α and γ should be least to the order of magnitude as the parameter ϵ.

To demonstrate the superior computational efficiency of the RCANGF method for solving the ground state of spin-1 BECs compared to the GFLM method proposed in the referenced paper [15],we adopt identical explicit and semi-implicit numerical schemes for comparative analysis.

4. Discretization of the reconfigured continuous normalized accelerated gradient flow

In this section, we present several numerical strategies to discretize the RCNAGF (3.12). Our primary focus is on temporal discretization strategies and providing an explicit discretization scheme (the leap-frog method) along with two types of semi-implicit discretization methods. A Fourier pseudospectral method [25] is utilized for spatial discretization in all cases.

Taking $\mathbf{\Phi}^n$ be a numerical approximation of $\mathbf{\Phi}(\cdot, t_n)$ with $t_n = n\tau$ and $\tau > 0$ is a given time step length. We denote $(H_1\phi_1, H_0\phi_0, H_{-1}\phi_{-1})^T$ with

$$\begin{aligned} H_1\phi_1 &:= \frac{\delta E}{\delta \phi_1}(\mathbf{\Phi}^n) = \left[-\frac{1}{2}\Delta + V(\mathbf{x}) + p_1(\mathbf{\Phi}^n)\right]\phi_1^n + \beta_s \bar{\phi}_{-1}^n(\phi_0^n)^2, \\ H_0\phi_0 &:= \frac{\delta E}{\delta \phi_0}(\mathbf{\Phi}^n) = \left[-\frac{1}{2}\Delta + V(\mathbf{x}) + p_0(\mathbf{\Phi}^n)\right]\phi_0^n + 2\beta_s \phi_{-1}^n\bar{\phi}_0^n\phi_1^n, \\ H_{-1}\phi_{-1} &:= \frac{\delta E}{\delta \phi_{-1}}(\mathbf{\Phi}^n) = \left[-\frac{1}{2}\Delta + V(\mathbf{x}) + p_{-1}(\mathbf{\Phi}^n)\right]\phi_{-1}^n + \beta_s (\phi_0^n)^2\bar{\phi}_1^n, \end{aligned} \tag{4.1}$$

where $p_1(\mathbf{\Phi}^n)$, $p_0(\mathbf{\Phi}^n)$, $p_{-1}(\mathbf{\Phi}^n)$ are defined as (2.3).

In the subsequent temporal discretizations for RCNAGF (3.12), the explicit Lagrange multiplier will be approximated by

$$\xi^0 = \frac{R_0D_0 - M_0F_0}{N_0R_0 - M_0^2}, \quad \xi^n = \frac{R_n[D_n - \|(\boldsymbol{\Phi}^n - \boldsymbol{\Phi}^{n-1})/\tau\|^2] - M_n[F_n - \|C(\boldsymbol{\Phi}^n - \boldsymbol{\Phi}^{n-1})/\tau\|^2]}{N_nR_n - M_n^2}, \quad n \neq 0, \tag{4.2}$$

and

$$\sigma^0 = \frac{N_0F_0 - M_0D_0}{N_0R_0 - M_0^2}, \quad \sigma^n = \frac{N_n[F_n - \|C(\boldsymbol{\Phi}^n - \boldsymbol{\Phi}^{n-1})/\tau\|^2] - M_n[D_n - \|(\boldsymbol{\Phi}^n - \boldsymbol{\Phi}^{n-1})/\tau\|^2]}{N_nR_n - M_n^2}, \quad n \neq 0, \tag{4.3}$$

where

$$\begin{cases} N_n = \|\phi_1^n\|^2 + \|\phi_0^n\|^2 + \|\phi_{-1}^n\|^2, \quad M_n = \|\phi_1^n\|^2 - \|\phi_{-1}^n\|^2, \quad R_n = \|\phi_1^n\|^2 + \|\phi_{-1}^n\|^2, \\ D^n = \int_\Omega (\mathbf{\Phi}^n)^* H(\mathbf{\Phi}^n)\mathbf{\Phi}^n d\mathbf{x} \quad F^n = \int_\Omega (\mathbf{\Phi}^n)^* C H(\mathbf{\Phi}^n)\mathbf{\Phi}^n d\mathbf{x}. \end{cases}$$

Here, we apply the forward first-order finite difference to approximate $\partial_t \mathbf{\Phi}$ in the Lagrange multipliers $\xi_{\mathbf{\Phi}}(t)$ and $\sigma_{\mathbf{\Phi}}(t)$.

4.1. Stabilized semi-implicit temporal discretization

We first consider the stabilized semi-implicit scheme for RCNAGF (RCNAGF-SI), which reads as follows:

$$\begin{cases} \dfrac{\mathbf{\Phi}^{n+1} - 2\mathbf{\Phi}^n + \mathbf{\Phi}^{n-1}}{\tau^2} + \dfrac{\eta}{n\tau}\dfrac{\mathbf{\Phi}^{n+1} - \mathbf{\Phi}^{n-1}}{2\tau} = \left[\dfrac{1}{2}\Delta - S^n\right]\boldsymbol{\Phi}^{n+1} + \left[\xi^n I_3 + \sigma^n C\right]\mathbf{\Phi}^n & (a) \\ \qquad + (S^n - V(\mathbf{x}) - A(\mathbf{\Phi}^n) - B(\mathbf{\Phi}^n))\mathbf{\Phi}^n + \alpha^n \mathbf{\Xi}(\mathbf{\Phi}^n) + \gamma^n \mathbf{\Upsilon}(\mathbf{\Phi}^n), & \\ N_{n+1} := \|\phi_1^{n+1}\|^2 + \|\phi_0^{n+1}\|^2 + \|\phi_{-1}^{n+1}\|^2 = 1, & (b) \\ M_{n+1} := \|\phi_1^{n+1}\|^2 - \|\phi_{-1}^{n+1}\|^2 = m, & (c) \\ \mathbf{\Phi}(\mathbf{x}, 0) = \mathbf{\Phi}^0(\mathbf{x}), \quad \partial_t\mathbf{\Phi}(\mathbf{x}, 0) = \mathbf{v}^0(\mathbf{x}) = \mathbf{0}, \quad \mathbf{x} \in \Omega, \quad M_{\boldsymbol{\Phi}^0} = m \quad \text{and} \quad N_{\boldsymbol{\Phi}^0} = 1, & (d) \end{cases} \tag{4.4}$$

where $\eta \geq 3$ is a damping coefficient and $S^n = \mathrm{diag}(s_1^n, s_0^n, s_{-1}^n)$ are the stabilization parameters. The Lagrange multipliers ξ^n and σ^n are defined as (4.2) and (4.3), respectively. The related supplied functions of α^n and γ^n are $\Xi(\mathbf{\Phi}) = H(\mathbf{\Phi})\mathbf{\Phi}$ and $\Upsilon(\mathbf{\Phi}) = (\phi_1, 0, -\phi_{-1})^T$, respectively. The reason we choose these supplied functions is to accurately compute the ground states of spin-1 BECs.

Our chosen termination condition is

$$\|\mathbf{\Phi}^{n+1} - \mathbf{\Phi}^n\|_\infty \leq \epsilon, \tag{4.5}$$

and when $\epsilon = 0$ (i.e. $\phi_1^{n+1} = \phi_1^n, \phi_0^{n+1} = \phi_0^n, \phi_{-1}^{n+1} = \phi_{-1}^n$), it indicates that our scheme has ceased its evolution. This also implies that

$$\begin{aligned} \mathbf{0} &= \left(\frac{1}{2}\Delta - V(\mathbf{x}) - A(\mathbf{\Phi}^n) - B(\mathbf{\Phi}^n)\right)\mathbf{\Phi}^n + \left[\xi^n I_3 + \sigma^n C\right]\mathbf{\Phi}^n + \alpha^n \mathbf{\Xi}(\mathbf{\Phi}^n) + \gamma^n \mathbf{\Upsilon}(\mathbf{\Phi}^n) \\ &= -H(\mathbf{\Phi}^n)\mathbf{\Phi}^n + \left[\mu^n I_3 + \lambda^n C\right]\mathbf{\Phi}^n + \alpha^n(H(\mathbf{\Phi}^n)\mathbf{\Phi}^n) + \gamma^n(\phi_1^n, 0, -\phi_{-1}^n)^T. \end{aligned} \tag{4.6}$$

After calculating each of the components of (4.6), it can be seen that

$$\begin{aligned} \left[-\frac{1}{2}\Delta + V(\mathbf{x}) + p_1(\mathbf{\Phi}^n)\right]\phi_1^n + \beta_s\bar{\phi}_{-1}^n(\phi_0^n)^2 &= \frac{\xi^n + \sigma^n + \gamma^n}{1 - \alpha^n}\phi_1 = \frac{\mu^n + \lambda^n + \gamma^n}{1 - \alpha^n}\phi_1, \\ \left[-\frac{1}{2}\Delta + V(\mathbf{x}) + p_0(\mathbf{\Phi}^n)\right]\phi_0^n + 2\beta_s\phi_{-1}^n\bar{\phi}_0^n\phi_1^n &= \frac{\xi^n}{1 - \alpha^n}\phi_0 = \frac{\mu^n}{1 - \alpha^n}\phi_0, \\ \left[-\frac{1}{2}\Delta + V(\mathbf{x}) + p_{-1}(\mathbf{\Phi}^n)\right]\phi_{-1}^n + \beta_s(\phi_0^n)^2\bar{\phi}_1^n &= \frac{\xi^n - \sigma^n - \gamma^n}{1 - \alpha^n}\phi_{-1} = \frac{\mu^n - \lambda^n - \gamma^n}{1 - \alpha^n}\phi_{-1}, \end{aligned} \tag{4.7}$$

with $p_1(\mathbf{\Phi}^n)$, $p_0(\mathbf{\Phi}^n)$, $p_{-1}(\mathbf{\Phi}^n)$ are defined as (2.3).

The eigenvalues of $\phi_1, \phi_0, \phi_{-1}$ are $(\mu^n+\lambda^n+\gamma^n)/(1-\alpha^n)$, $\mu^n/(1-\alpha^n)$, $(\mu^n-\lambda^n-\gamma^n)/(1-\alpha^n)$, respectively. We can observe that the eigenvalues ϕ_1 ϕ_0, and ϕ_{-1} satisfy the relation

$$\frac{\mu^n+\lambda^n+\gamma^n}{1-\alpha^n}+\frac{\mu^n-\lambda^n-\gamma^n}{1-\alpha^n}=\frac{2\mu^n}{1-\alpha^n}\to \mu_1+\mu_{-1}=2\mu_0.$$

In conclusion, our scheme (4.4) can match the Euler–Lagrange Eqs. (2.9).

As this is a three-level scheme, for initialization, we calculate the initial step $\boldsymbol{\Phi}_1 \approx \boldsymbol{\Phi}_1(\cdot,\tau)$ by using the second-order Taylor's expansion at $t=0$ and noting $\partial_t \mathbf{\Phi}(\cdot,0)=\mathbf{v}_0=\mathbf{0}$:

$$\mathbf{\Phi}^1=\mathbf{\Phi}^0+\tau\partial_t\mathbf{\Phi}|_{t=0}+\frac{\tau^2}{2}\partial_{tt}\mathbf{\Phi}|_{t=0}=\mathbf{\Phi}^0+\frac{\tau^2}{2}\left(-H(\mathbf{\Phi}^0)+(\xi_0 I_3+\sigma_0 C)\mathbf{\Phi}^0\right). \tag{4.8}$$

It is worth mentioning that in RCNAGF-SI (4.4), we only need to solve nine linear elliptic equations with constant coefficients along with an additional low-dimensional algebraic system at each time step. Therefore, RCNAGF-SI (4.4) is highly efficient, especially when fast Poisson solvers are available. We now demonstrate that this scheme can be effectively implemented.

Algorithm 1 RCNAGF-SI

Give the initial value $\mathbf{\Phi}^0=(\phi_1^0,\phi_0^0,\phi_{-1}^0)^T$, time step $\tau$and damping coefficient η;
Compute $\mathbf{\Phi}^1$ according to equation (4.8) and set $\boldsymbol{\Phi}^{n+1}=\left(\mathbf{D}^{n+1}+\alpha^n\mathbf{D}_1^{n+1}+\gamma^n\mathbf{D}_2^{n+1}\right)$;
while not converge **do**
  Compute $H(\mathbf{\Phi}^n)$, ξ^n and σ^n according to (4.1), (4.2) and (4.3), respectively.
  Take $\boldsymbol{\Phi}^{n+1}=\left(\mathbf{D}^{n+1}+\alpha^n\mathbf{D}_1^{n+1}+\gamma^n\mathbf{D}_2^{n+1}\right)$ in the equation (4.4)(a), we have

$$L\left(\mathbf{D}^{n+1}+\alpha^n\mathbf{D}_1^{n+1}+\gamma^n\mathbf{D}_2^{n+1}\right)=\mathbf{W}^n+\alpha^n\mathbf{W}_1^n+\gamma^n\mathbf{W}_2^n,$$

  where $L:=\left(1+\frac{\tau}{2}\frac{\eta}{n\tau}+S^n-\frac{\tau^2}{2}\Delta\right)$ and

$$\begin{cases}\mathbf{W}^n=\left(-1+\frac{\tau}{2}\frac{\eta}{n\tau}\right)\mathbf{\Phi}^{n-1}+2\mathbf{\Phi}^n+\tau^2\left(S^n+\xi^n I_3+\sigma^n C-H(\mathbf{\Phi}^n)-\frac{1}{2}\Delta\right)\mathbf{\Phi}^n,\\ \mathbf{W}_1^n=\tau^2\mathbf{\Xi}(\mathbf{\Phi}^n),\quad \mathbf{W}_2^n=\tau^2\mathbf{\Upsilon}(\mathbf{\Phi}^n).\end{cases}$$

  Solve the linear elliptic equation for $\left(\mathbf{D}^{n+1},\mathbf{D}_1^{n+1},\mathbf{D}_2^{n+1}\right)$

$$L\mathbf{D}^{n+1}=\mathbf{W}^n,\quad L\mathbf{D}_1^{n+1}=\mathbf{W}_1^n,\quad L\mathbf{D}_2^{n+1}=\mathbf{W}_2^n.$$

  Solve a low-dimensional algebraic system for α^n and γ^n by

$$\begin{cases}N_{n+1}=:\|\mathbf{\Phi}^{n+1}\|^2=\|\phi_1^{n+1}\|^2+\|\phi_0^{n+1}\|^2+\|\phi_{-1}^{n+1}\|^2=1,\\ M_{n+1}:=\|\phi_1^{n+1}\|^2-\|\phi_{-1}^{n+1}\|^2=m.\end{cases} \tag{4.9}$$

  Update $\mathbf{\Phi}^{n+1}=L^{-1}\left(\mathbf{W}^n+\alpha^n\mathbf{W}_1^n+\gamma^n\mathbf{W}_2^n\right)$.
  n:= n+1.
end while

It should be noted that when RCNAGF-SI (4.4) is fully discretized, the linear operator L is invertible. Due to the nonlinearity of the system (4.9), it is generally challenging to prove its unique solvability unconditionally. Numerical experiments have shown that obtaining a numerical solution is not difficult as long as the time step size is small. Therefore, in cases where the system is solvable, effective iterative methods such as Newton iteration can be used to solve it. To achieve the desired solutions for α^n and γ^n to be zero, we can set the initial values to 0 in the Newton iteration, which typically leads to a fast convergence.

Remark 4.1. The scheme (4.4) can be extended to compute the arbitrary multicomponent BEC ground state, which can be modeled as a continuous normalized gradient flow.

Remark 4.2. The linear implicit part $\left[\frac{1}{2}\Delta-S^n\right]\boldsymbol{\Phi}^{n+1}$ can also be replaced by $\left[\frac{1}{2}\Delta-S^n\right]\frac{\boldsymbol{\Phi}^{n+1}-\boldsymbol{\Phi}^{n-1}}{2}$, which does not affect the computational results.

4.2. Leap-frog temporal discretization

We then implement an explicit leap-frog scheme (RCNAGF-LF) for system (3.12), which takes the following form:

$$\begin{cases} \dfrac{\mathbf{\Phi}^{n+1}-2\mathbf{\Phi}^{n}+\mathbf{\Phi}^{n-1}}{\tau^2}+\dfrac{\eta}{n\tau}\dfrac{\mathbf{\Phi}^{n+1}-\mathbf{\Phi}^{n-1}}{2\tau}=-H(\mathbf{\Phi}^n)\mathbf{\Phi}^n+\left[\xi^n I_3+\sigma^n C\right]\mathbf{\Phi}^n & (a)\\ \quad +\alpha^n\mathbf{\Xi}(\mathbf{\Phi}^n)+\gamma^n\mathbf{\Upsilon}(\mathbf{\Phi}^n), & \\ N_{n+1}:=\|\phi_1^{n+1}\|^2+\|\phi_0^{n+1}\|^2+\|\phi_{-1}^{n+1}\|^2=1, & (b)\\ M_{n+1}:=\|\phi_1^{n+1}\|^2-\|\phi_{-1}^{n+1}\|^2=m, & (c)\\ \mathbf{\Phi}(\mathbf{x},0)=\mathbf{\Phi}^0(\mathbf{x}),\quad \partial_t\mathbf{\Phi}(\mathbf{x},0)=\mathbf{v}^0(\mathbf{x})=\mathbf{0},\quad \mathbf{x}\in\Omega,\quad M_{\mathbf{\Phi}^0}=m\quad\text{and}\quad N_{\mathbf{\Phi}^0}=1, & (d)\end{cases} \tag{4.10}$$

where $\eta \geq 3$ is a damping coefficient, ξ^n and σ^n are defined as (4.2) and (4.3), respectively. The related supplementary functions of α^n and γ^n are $\Xi(\mathbf{\Phi}) = H(\mathbf{\Phi})\mathbf{\Phi}$ and $\Upsilon(\mathbf{\Phi}) = (\phi_1, 0, -\phi_{-1})^T$, respectively. Using second-order Taylor expansion at $t = 0$ with vanishing initial velocity $\partial_t\mathbf{\Phi}(\cdot, 0) = \mathbf{0}$, we obtain $\mathbf{\Phi}^1$ as specified in (4.8).

The stopping criterion for RCNAGF-LF (4.10) is defined as (4.5). When the equation ((4.10)(a)) stops evolution, we have

$$\begin{aligned}\mathbf{0}&=-H(\mathbf{\Phi}^n)\mathbf{\Phi}^n+\left[\xi^n I_3+\sigma^n C\right]\mathbf{\Phi}^n+\alpha^n\mathbf{\Xi}(\mathbf{\Phi}^n)+\gamma^n\mathbf{\Upsilon}(\mathbf{\Phi}^n)\\ &=-H(\mathbf{\Phi}^n)\mathbf{\Phi}^n+\left[\mu^n I_3+\lambda^n C\right]\mathbf{\Phi}^n+\alpha^n(H(\mathbf{\Phi}^n)\mathbf{\Phi}^n)+\gamma^n(\phi_1^n,0,-\phi_{-1}^n)^T.\end{aligned} \tag{4.11}$$

After calculating each of the components of (4.11), it can be seen that

$$\begin{aligned}&\left[-\frac{1}{2}\Delta+V(\mathbf{x})+p_1(\mathbf{\Phi}^n)\right]\phi_1^n+\beta_s\bar{\phi}_{-1}^n(\phi_0^n)^2=\frac{\xi^n+\sigma^n+\gamma^n}{1-\alpha^n}\phi_1=\frac{\mu^n+\lambda^n+\gamma^n}{1-\alpha^n}\phi_1,\\ &\left[-\frac{1}{2}\Delta+V(\mathbf{x})+p_0(\mathbf{\Phi}^n)\right]\phi_0^n+2\beta_s\phi_{-1}^n\bar{\phi}_0^n\phi_1^n=\frac{\xi^n}{1-\alpha^n}\phi_0=\frac{\mu^n}{1-\alpha^n}\phi_0,\\ &\left[-\frac{1}{2}\Delta+V(\mathbf{x})+p_{-1}(\mathbf{\Phi}^n)\right]\phi_{-1}^n+\beta_s(\phi_0^n)^2\bar{\phi}_1^n=\frac{\xi^n-\sigma^n-\gamma^n}{1-\alpha^n}\phi_{-1}=\frac{\mu^n-\lambda^n-\gamma^n}{1-\alpha^n}\phi_{-1},\end{aligned} \tag{4.12}$$

with $p_1(\mathbf{\Phi}^n)$, $p_0(\mathbf{\Phi}^n)$, $p_{-1}(\mathbf{\Phi}^n)$ are defined as (2.3).

The eigenvalues of $\phi_1, \phi_0, \phi_{-1}$ are $(\mu^n + \lambda^n + \gamma^n)/(1 - \alpha^n)$, $\mu^n/(1 - \alpha^n)$, $(\mu^n - \lambda^n - \gamma^n)/(1 - \alpha^n)$, respectively. We can observe that the eigenvalues ϕ_1 ϕ_0, and ϕ_{-1} satisfy the relation

$$\frac{\mu^n+\lambda^n+\gamma^n}{1-\alpha^n}+\frac{\mu^n-\lambda^n-\gamma^n}{1-\alpha^n}=\frac{2\mu^n}{1-\alpha^n}\rightarrow\mu_1+\mu_{-1}=2\mu_0.$$

In conclusion, our scheme can match the Euler–Lagrange Eqs. (2.9).

In detail, the RCNAGF-LF (4.10) can be summarized as in Algorithm 2.

Algorithm 2 RCNAGF-LF

Give the initial value $\mathbf{\Phi}^0 = (\phi_1^0, \phi_0^0, \phi_{-1}^0)^T$, time step $\tau$and damping coefficient η;
Compute $\mathbf{\Phi}^1$ according to equation (4.8) and set $\Phi^{n+1} = \left(\mathbf{D}^{n+1} + \alpha^n\mathbf{D}_1^{n+1} + \gamma^n\mathbf{D}_2^{n+1}\right)$;
while not converge **do**
  Compute $H(\mathbf{\Phi}^n)$, ξ^n and σ^n according to (4.1), (4.2) and (4.3), respectively.

  Take $\Phi^{n+1} = \left(\mathbf{D}^{n+1} + \alpha^n\mathbf{D}_1^{n+1} + \gamma^n\mathbf{D}_2^{n+1}\right)$ in the equation (4.10)(a), we have

$$\mathbf{D}^{n+1}=\frac{(\tau\frac{\eta}{n\tau}-2)\mathbf{\Phi}^{n-1}+4\mathbf{\Phi}^n+2\tau^2(\xi^n I_3+\sigma^n C-H(\mathbf{\Phi}^n))\mathbf{\Phi}^n}{2+\tau\frac{\eta}{n\tau}},$$

$$\mathbf{D}_1^{n+1}=\frac{2\tau^2\mathbf{\Xi}(\mathbf{\Phi}^n)}{2+\tau\frac{\eta}{n\tau}}\quad\textit{and}\quad\mathbf{D}_2^{n+1}=\frac{2\tau^2\mathbf{\Upsilon}(\mathbf{\Phi}^n)}{2+\tau\frac{\eta}{n\tau}}.$$

  Solve a low-dimensional algebraic system for α^n and γ^n by

$$\begin{cases} N_{n+1}=:\|\mathbf{\Phi}^{n+1}\|^2=\|\phi_1^{n+1}\|^2+\|\phi_0^{n+1}\|^2+\|\phi_{-1}^{n+1}\|^2=1,\\ M_{n+1}:=\|\phi_1^{n+1}\|^2-\|\phi_{-1}^{n+1}\|^2=m.\end{cases}$$

  Update $\mathbf{\Phi}^{n+1} = \mathbf{D}^{n+1} + \alpha^n\mathbf{D}_1^{n+1} + \gamma^n\mathbf{D}_2^{n+1}$.
  n:= n+1.
end while

4.3. Second order stabilized semi-implicit temporal discretization

The main challenge in constructing a second-order semi-implicit scheme for RCNAGF (3.12) is the explicit approximation of the first-order derivative terms $\partial_t\mathbf{\Phi}$ in the Lagrange multiplier terms $\xi_{\mathbf{\Phi}}(t)$ and $\sigma_{\mathbf{\Phi}}(t)$. Specifically, to achieve second-order accuracy for the Lagrange multipliers $\xi_{\mathbf{\Phi}}(t)$ and $\sigma_{\mathbf{\Phi}}(t)$, a four-step scheme is required. However, initial conditions are missing to initialize this scheme. To address this, we first compute the value at the first step $\mathbf{\Phi}^1$ using a second-order Taylor expansion of $\mathbf{\Phi}(\cdot,0)$ around $t=0$. After defining the RCNAGF-SI scheme, we use it to compute the values at the second step $\mathbf{\Phi}^2$. This approach enables the efficient construction of a second-order scheme. The specific form of the scheme (RCNAGF-2SI) is given below:

$$\begin{cases} \dfrac{\mathbf{\Phi}^{n+1}-2\mathbf{\Phi}^n+\mathbf{\Phi}^{n-1}}{\tau^2}+\dfrac{\eta}{n\tau}\dfrac{\mathbf{\Phi}^{n+1}-\mathbf{\Phi}^{n-1}}{2\tau}=\left[\dfrac{1}{2}\Delta-S^n\right]\mathbf{\Phi}^{n+1}+\left[\xi^n I_3+\sigma^n C\right]\mathbf{\Phi}^{n,*} & (a)\\ \quad+\left(S^n-V(\mathbf{x})-A(\mathbf{\Phi}^{n,*})-B(\mathbf{\Phi}^{n,*})\right)\mathbf{\Phi}^{n,*}+\alpha^n\mathbf{\Xi}(\mathbf{\Phi}^{n,*})+\gamma^n\mathbf{\Upsilon}(\mathbf{\Phi}^{n,*}), & \\ N_{n+1}=\|\phi_1^{n+1}\|^2+\|\phi_0^{n+1}\|^2+\|\phi_{-1}^{n+1}\|^2=1, & (b)\\ M_{n+1}=\|\phi_1^{n+1}\|^2-\|\phi_{-1}^{n+1}\|^2=m, & (c)\\ \mathbf{\Phi}(\mathbf{x},0)=\mathbf{\Phi}^0(\mathbf{x}),\quad \partial_t\mathbf{\Phi}(\mathbf{x},0)=\mathbf{v}^0(\mathbf{x})=\mathbf{0},\quad \mathbf{x}\in\Omega,\quad M_{\mathbf{\Phi}^0}=m\quad\text{and}\quad N_{\mathbf{\Phi}^0}=1, & (d) \end{cases} \tag{4.13}$$

where $\mathbf{\Phi}^{n,*}=\frac{3}{2}\mathbf{\Phi}^n-\frac{1}{2}\mathbf{\Phi}^{n-1}$, $\eta\geq 3$ is a damping coefficient and $S=\mathrm{diag}(s_1^n,s_0^n,s_{-1}^n)$ are the stabilization parameters. The related supplementary functions of α^n and γ^n are $\Xi(\mathbf{\Phi})=H(\mathbf{\Phi})\mathbf{\Phi}$ and $\Upsilon(\mathbf{\Phi})=(\phi_1,0,-\phi_{-1})^T$, respectively. The Lagrange multipliers ξ^n and σ^n can be obtained as follows:

$$\begin{cases} \xi^n=\frac{R_n[D_n-\|(\mathbf{\Phi}^n-\mathbf{\Phi}^{n-2})/(2\tau)\|^2]-M_n[F_n-\|C(\mathbf{\Phi}^n-\mathbf{\Phi}^{n-2})/(2\tau)\|^2]}{N_nR_n-M_n^2}, & n\geq 3,\\ \sigma^n=\frac{N_n[F_n-\|C(\mathbf{\Phi}^n-\mathbf{\Phi}^{n-2})/(2\tau)\|^2]-M_n[D_n-\|(\mathbf{\Phi}^n-\mathbf{\Phi}^{n-2})/(2\tau)\|^2]}{N_nR_n-M_n^2}, & n\geq 3, \end{cases} \tag{4.14}$$

where

$$\begin{cases} N_n=\|\phi_1^{n,*}\|^2+\|\phi_0^{n,*}\|^2+\|\phi_{-1}^{n,*}\|^2 \quad, M_n=\|\phi_1^{n,*}\|^2-\|\phi_{-1}^{n,*}\|^2,\quad R_n=\|\phi_1^{n,*}\|^2+\|\phi_{-1}^{n,*}\|^2,\\ D^n=\int_\Omega(\mathbf{\Phi}^{n,*})^*H(\mathbf{\Phi}^{n,*})\mathbf{\Phi}^{n,*}d\mathbf{x}\quad F^n=\int_\Omega(\mathbf{\Phi}^{n,*})^*CH(\mathbf{\Phi}^{n,*})\mathbf{\Phi}^{n,*}d\mathbf{x}. \end{cases}$$

The stopping criterion for RCNAGF-2SI (4.13) is defined as (4.5). When the scheme (4.13) stops evolution, we have $\mathbf{\Phi}^{n+1}=\mathbf{\Phi}^n=\mathbf{\Phi}^{n-1}=\mathbf{\Phi}^{n,*}$. It is evident that the final evolution equation of ((4.13),(a)) can accurately match the Euler–Lagrange Eq. (2.9). Since

$$\begin{aligned} \mathbf{0}&=\left(\frac{1}{2}\Delta-V(\mathbf{x})-A(\mathbf{\Phi}^{n,*})-B(\mathbf{\Phi}^{n,*})\right)\mathbf{\Phi}^{n,*}+\left[\xi^nI_3+\sigma^nC\right]\mathbf{\Phi}^{n,*}+\alpha^n\mathbf{\Xi}(\mathbf{\Phi}^{n,*})+\gamma^n\mathbf{\Upsilon}(\mathbf{\Phi}^{n,*})\\ &=-H(\mathbf{\Phi}^n)\mathbf{\Phi}^n+\left[\mu^nI_3+\lambda^nC\right]\mathbf{\Phi}^n+\alpha^n(H(\mathbf{\Phi}^n)\mathbf{\Phi}^n)+\gamma^n(\phi_1^n,0,-\phi_{-1}^n)^T, \end{aligned} \tag{4.15}$$

and the remaining proof process is similar to the previous one and will be omitted appropriately.

5. Numerical results for spin-1 BECs

In this section, we provide several numerical examples to demonstrate the accuracy and efficiency of the proposed schemes.

We choose the same damping coefficient η in the RCNAGF-SI, RCNAGF-LF, and RCNAGF-2SI. The parameter η can be adjusted for optimal numerical performance. Furthermore, the stabilization parameters S^n in all semi-implicit schemes are chosen as the following form:

$$\theta\max\left(V(\mathbf{x})+A(\mathbf{\Phi}^n)+B(\mathbf{\Phi}^n)\right),\quad \theta\in[0,1]. \tag{5.1}$$

Finally, the stopping criterion is $\|\mathbf{\Phi}^{n+1}-\mathbf{\Phi}^n\|_\infty\leq\epsilon$ with $\epsilon=10^{-12}$ in the absence of specifying alternative stopping criteria. For fair comparisons, we conduct all numerical experiments on the bounded domain Ω with periodic boundary conditions using the Fourier pseudospectral method described in [25]. To quantify the ground state solution $\mathbf{\Phi}_g(\mathbf{x})$, we define the residual (measured at the grid points) for the Euler–Lagrange Eq. (2.9)

$$maxres=\left\|(\xi^nI_3+\sigma^nC)\mathbf{\Phi}^n-\left[-\frac{1}{2}\Delta+V(\mathbf{x})+A(\mathbf{\Phi}^n)+B(\mathbf{\Phi}^n)\right]\mathbf{\Phi}^n\right\|_\infty.$$

All experiments are conducted on a workstation with a 4.40 GHz CPU, using Matlab (2023b). Below, we used "CPU(s)" to indicate the CPU time required for convergence.

5.1. Numerical comparison

In this subsection, we perform numerical comparisons of our schemes proposed in Section 4 (RCNAGF-SI, RCNAGF-2SI, RCANGF-LF) with other schemes based on gradient flows, including GFLM-BF [15] and RCNGF-BF [38], GFLM-FE [15].

Example 5.1. In this example, we investigate the properties of our schemes through a simple example for the magnetization $m=0.5$ in 1D with harmonic potential $V(x)=\frac{x^2}{2}+25sin^2(\frac{\pi x}{4})$ for the spatial step $h=1/16$ in the domain $\Omega=[-16,16]$, and consider the following two cases [15]:

Case I. (ferromagnetic condensates, e.g., 87Rb) $\beta_n = 885$ and $\beta_s = -4.1$, and the initial data is $\Phi_0(\mathbf{x}) = \left(\frac{\sqrt{1+3M}}{2},\sqrt{\frac{1-M}{2}},\frac{\sqrt{1-M}}{2}\right)\phi_g^{ap}(x)$ with $\phi_g^{ap}(x)=e^{-x^2/2}/\pi^{1/4}$.

Case II. (antiferromagnetic condensates, e.g., 23Na) $\beta_n = 241$ and $\beta_s = 7.5$, and the initial data is $\Phi_0(\mathbf{x}) = \left(\sqrt{\frac{1+M}{2}},0,\sqrt{\frac{1-M}{2}}\right)\phi_g^{ap}(x)$ with $\phi_g^{ap}(x)=e^{-x^2/2}/\pi^{1/4}$.

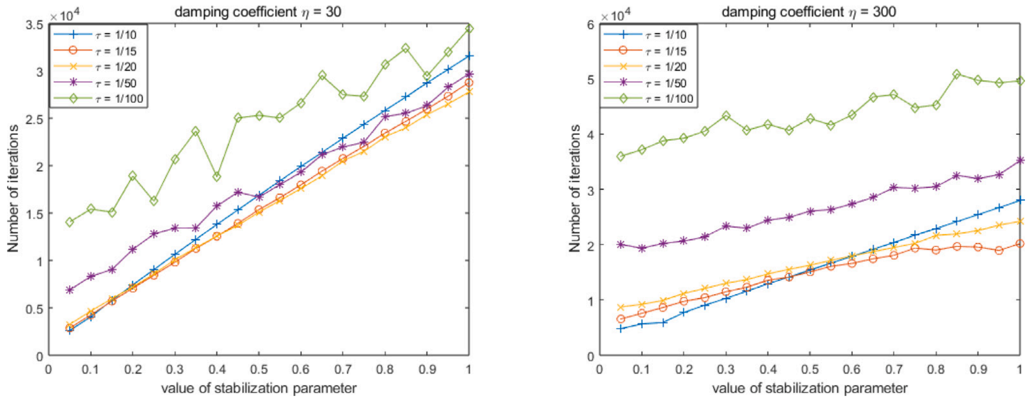

Fig. 1. Different stability parameters θ affect the RCNAGF-SI in Example 5.1 for Case I (Left: damping coefficient $\eta = 30$; Right: damping coefficient $\eta = 300$).

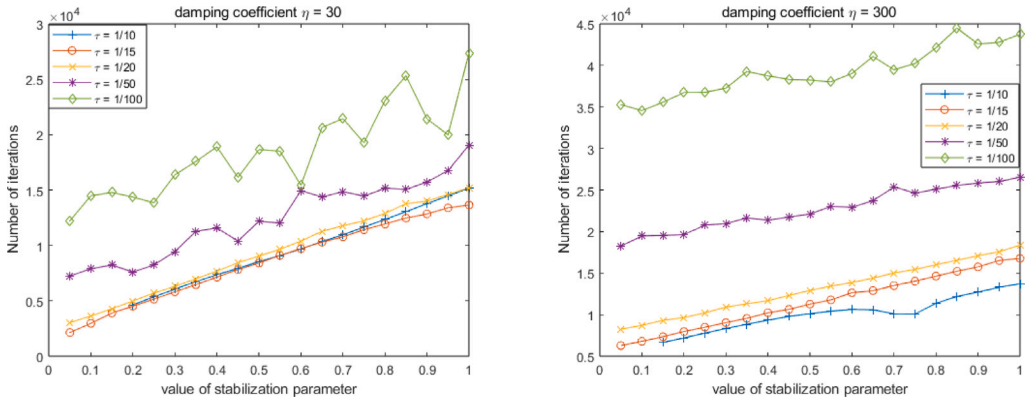

Fig. 2. Different stability parameters θ affect the RCNAGF-2SI in Example 5.1 for Case I (Left: damping coefficient $\eta = 30$; Right: damping coefficient $\eta = 300$).

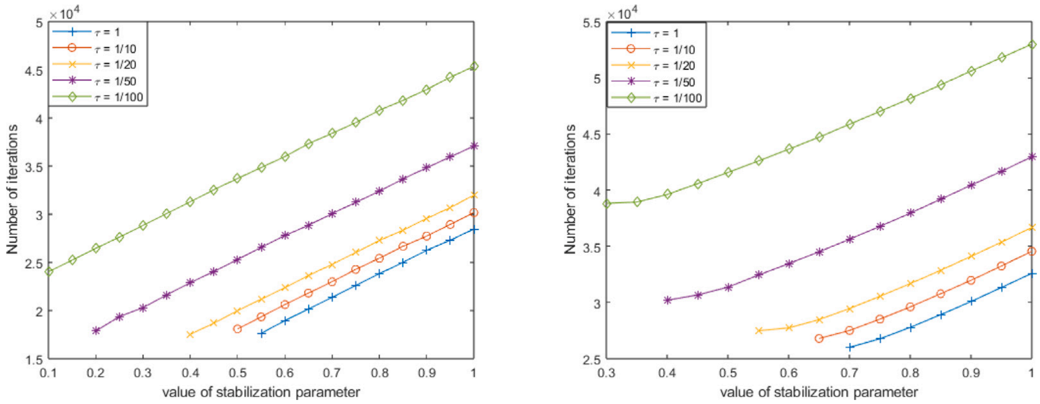

Fig. 3. Different stability parameters θ affect in Example 5.1 for Case I (Left: GFLM-BF; Right: RCNGF-BF).

Before comparing the efficiency of different schemes, let us first test the impact of parameters in the semi-implicit schemes on the computational results from Case I.

From Figs. 1–3, it can be seen that the number of iterations required to compute the ground states of spin-1 BECs with semi-implicit schemes increases as the stability parameter θ increases at the same time step size. With a shorter time step of τ, the number of iterations increases, but at the same time, the required stability parameter θ decreases. As a result, the variation in the number of iterations becomes more complex. Additionally, as shown in Fig. 4, we can see that as the stability parameter θ is fixed, the number of iterations required to compute the ground states of spin-1 BECs for RCNAGF-SI and RCNAGF-2SI increases as the damping parameter η increases.

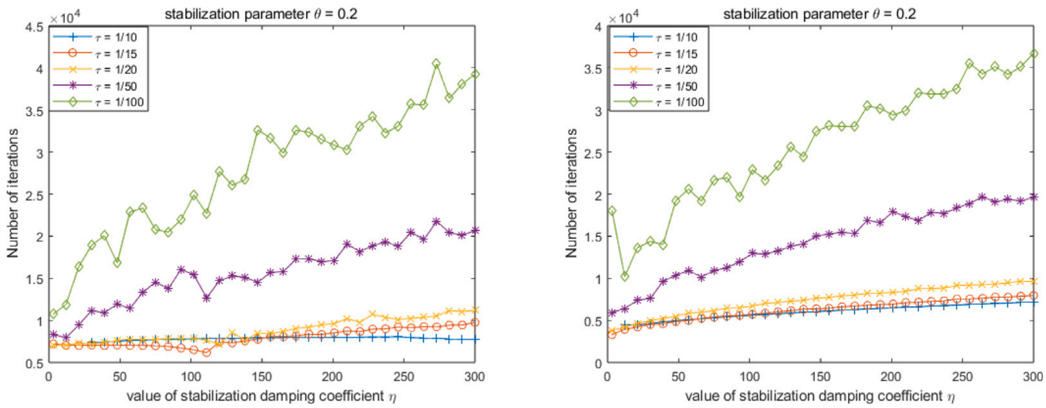

Fig. 4. Different damping coefficient η affect in Example 5.1 for Case I (Left: RCNAGF-SI; Right: RCNAGF-SI2).

Table 1
Numerical results in Example 5.1 (Case I) by different numerical schemes.

Method	τ	$E(\mathbf{\Phi}_g)$	ξ_g	σ_g	maxres	Iterations	CPU(s)	θ	η
GFLM-BF	1	47.6941680392	73.0222344821	0.0000000000	$7E{-}10^{-11}$	17 631	2.226432	0.55	–
	1/10	47.6941680392	73.0222344821	0.0000000000	$7E{-}10^{-11}$	18 083	2.363025	0.5	
	1/20	47.6941680392	73.0222344821	0.0000000000	$7E{-}10^{-11}$	17 525	2.348814	0.4	
	1/50	47.6941680392	73.0222344821	0.0000000000	$7E{-}10^{-11}$	17 865	2.265835	0.2	
	1/100	47.6941680392	73.0222344821	0.0000000000	$1E{-}10^{-10}$	24 031	2.996035	0.1	
RCNGF-BF	1	47.6941680392	73.0222344821	0.0000000000	$9E{-}10^{-11}$	26 008	4.741978	0.7	–
	1/10	47.6941680392	73.0222344821	0.0000000000	$9E{-}10^{-11}$	26 796	5.094073	0.65	
	1/20	47.6941680392	73.0222344821	0.0000000000	$9E{-}10^{-11}$	27 482	5.204191	0.55	
	1/50	47.6941680392	73.0222344821	0.0000000000	$9E{-}10^{-11}$	30 203	4.991901	0.4	
	1/100	47.6941680392	73.0222344821	0.0000000000	$1E{-}10^{-10}$	38 819	7.273091	0.3	
RCNAGF-SI	1/10	47.6941680392	73.0222344821	0.0000000000	$6E{-}10^{-12}$	2610	0.774786	0.05	30
	1/15	47.6941680392	73.0222344821	0.0000000000	$4E{-}10^{-12}$	1630	0.427671	0.05	3
	1/20	47.6941680392	73.0222344822	0.0000000000	$2E{-}10^{-11}$	2072	0.587890	0.05	3
	1/50	47.6941680392	73.0222344825	0.0000000000	$5E{-}10^{-10}$	5452	1.421419	0.05	9
	1/100	47.6941680392	73.0222344781	0.0000000000	$7E{-}10^{-10}$	10 707	2.321816	0.1	9
RCNAGF-2SI	1/10	47.6941680392	73.0222344821	0.0000000000	$1E{-}10^{-11}$	4429	0.774786	0.2	18
	1/15	47.6941680392	73.0222344821	0.0000000000	$2E{-}10^{-12}$	1385	0.427671	0.1	3
	1/20	47.6941680392	73.0222344821	0.0000000000	$5E{-}10^{-12}$	2292	0.587890	0.1	3
	1/50	47.6941680392	73.0222344819	0.0000000000	$2E{-}10^{-10}$	5927	1.421419	0.2	3
	1/100	47.6941680392	73.0222344821	0.0000000000	$2E{-}10^{-10}$	10 984	2.321816	0.25	21
GFLM-FE	1/800	47.6941680392	73.0222344821	0.0000000000	$7E{-}10^{-10}$	119 310	14.208451	–	–
	1/1000	47.6941680392	73.0222344821	0.0000000000	$9E{-}10^{-10}$	147 749	17.854081		
	1/2000	47.6941680392	73.0222344819	0.0000000000	$1E{-}10^{-9}$	279 694	33.131693		
RCNAGF-LF	1/100	47.6941680392	73.0222344821	0.0000000000	$6E{-}10^{-11}$	17 591	2.447742	–	3
	1/200	47.6941680392	73.0222344816	0.0000000000	$2E{-}10^{-10}$	19 339	2.713423		3
	1/400	47.6941680392	73.0222344815	0.0000000000	$5E{-}10^{-9}$	67 024	9.519665		3

To summarize, the speed at which the semi-implicit schemes RCANGF-SI and RCNAGF-2SI compute the ground states of spin-1 BECs is influenced by three parameters: the time step τ, the size of the stability parameter θ, and the damping parameter η. Generally, when an appropriate time step is chosen, such as $\tau = 1/20$, smaller stability and damping parameters lead to faster convergence of the scheme to the ground state.

Next, we provide two tables to compare the efficiency of our proposed schemes with those of gradient flow-based schemes, where the parameters are set to minimize the number of iterations required to reach the termination condition.

In Tables 1 and 2, we can see that our semi-implicit schemes have a clear advantage in terms of high rigidity, which only requires 1/8 of the total number of iteration steps in time step $0.05 \leq \tau \leq 0.1$ compared to the numerical schemes based on the normalized flow. When the solution system's rigidity is small, we only need 1/4 of the total number of iteration steps between $0.067 \leq \tau \leq 0.2$ compared to the numerical schemes based on the normalized flow. For explicit schemes, the maximum stable time step for GFLM-FE is approximately $1/800$, while RCNAGF-LF can choose a maximum of $1/100$.

Finally, it is evident that the Euler–Lagrange Eq. (2.9) has the maximum residual at the numerical solution derived through our schemes are of comparable magnitude to the tolerance ϵ from Table 1 and 2. This implies that our proposed schemes are capable of accurately computing the ground state of spin-1 BECs. We use Case II to verify if the numerical performance of our algorithm aligns with our theoretical analysis. In this scenario, the time step τ for all schemes is set to 1/15, the stabilization parameter

Table 2
Numerical results in Example 5.1 (Case II) by different numerical schemes.

Method	τ	$E(\mathbf{\Phi}_g)$	ξ_g	σ_g	maxres	Iterations	CPU(s)	θ	η
GFLM-BF	1	25.7289899366	37.3729555325	0.3461105978	4E−10^{-11}	1529	0.171645	0.35	–
	1/5	25.7289899366	37.3729555325	0.3461105978	4E−10^{-11}	1651	0.188564	0.35	
	1/10	25.7289899366	37.3729555325	0.3461105978	4E−10^{-11}	1608	0.183771	0.30	
	1/20	25.7289899366	37.3729555325	0.3461105978	4E−10^{-11}	1522	0.165727	0.20	
	1/100	25.7289899366	37.3729555322	0.3461105980	1E−10^{-10}	3131	0.348573	0.00	
RCNGF-BF	1	25.7289899366	37.3729555325	0.3461105978	4E−10^{-11}	1530	0.273201	0.35	–
	1/5	25.7289899366	37.3729555325	0.3461105978	5E−10^{-11}	1650	0.295272	0.35	
	1/10	25.7289899366	37.3729555322	0.3461105978	4E−10^{-11}	1608	0.292607	0.30	
	1/20	25.7289899366	37.3729555325	0.3461105978	4E−10^{-11}	1519	0.269971	0.20	
	1/100	25.7289899366	37.3729555325	0.3461105980	1E−10^{-10}	3131	0.564294	0.00	
RCNAGF-SI	1/5	25.7289899366	37.3729555328	0.3461105977	4E−10^{-11}	375	0.075305	0.05	24
	1/10	25.7289899366	37.3729555327	0.3461105977	5E−10^{-12}	414	0.084121	0.05	15
	1/15	25.7289899366	37.3729555328	0.3461105977	8E−10^{-11}	649	0.127240	0.1	15
	1/20	25.7289899366	37.3729555328	0.3461105977	7E−10^{-11}	859	0.165202	0.15	12
	1/100	25.7289899366	37.3729555327	0.3461105977	3E−10^{-11}	4729	1.099916	0.45	30
RCNAGF-2SI	1/5	25.7289899366	37.3729555327	0.3461105977	1E−10^{-12}	470	0.106634	0.2	3
	1/10	25.7289899366	37.3729555328	0.3461105977	5E−10^{-12}	402	0.082004	0.15	3
	1/15	25.7289899366	37.3729555323	0.3461105979	2E−10^{-11}	578	0.124971	0.15	15
	1/20	25.7289899366	37.3729555327	0.3461105977	4E−10^{-11}	791	0.169883	0.25	12
	1/100	25.7289899366	37.3729555324	0.3461105979	3E−10^{-11}	4254	0.799422	0.45	30
GFLM-FE	1/800	25.7289899366	37.3729555281	0.3461105977	7E−10^{-10}	21 248	2.209227	–	–
	1/1000	25.7289899366	37.3729555269	0.3461106003	9E−10^{-10}	26 077	2.815604		
	1/2000	25.7289899366	37.3729555211	0.3461106030	1E−10^{-9}	49 164	5.145951		
RCNAGF-LF	1/50	25.7289899366	37.3729555329	0.3461105977	2E−10^{-10}	2740	0.371658	–	3
	1/100	25.7289899366	37.3729555334	0.3461105976	6E−10^{-10}	9447	1.208115		3
	1/200	25.7289899366	37.3729555329	0.3461105977	1E−10^{-9}	33 186	4.123618		3

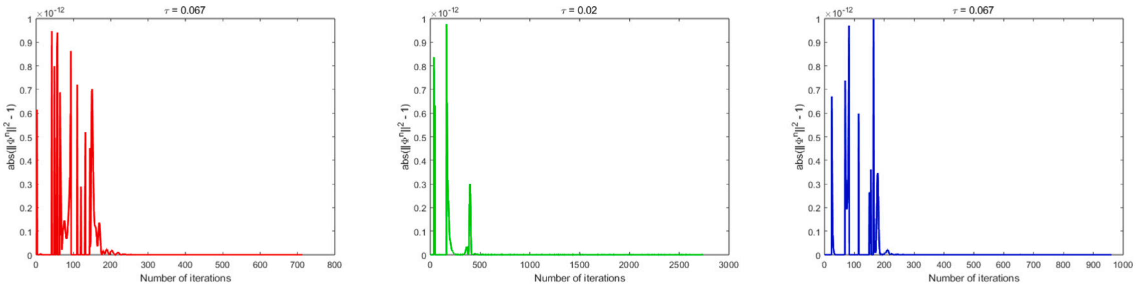

Fig. 5. Conservation of mass for Example 5.1 in Case II (Left: RCNAGF-SI; Middle: RCNAGF-LF; Right: RCNAGF-2SI).

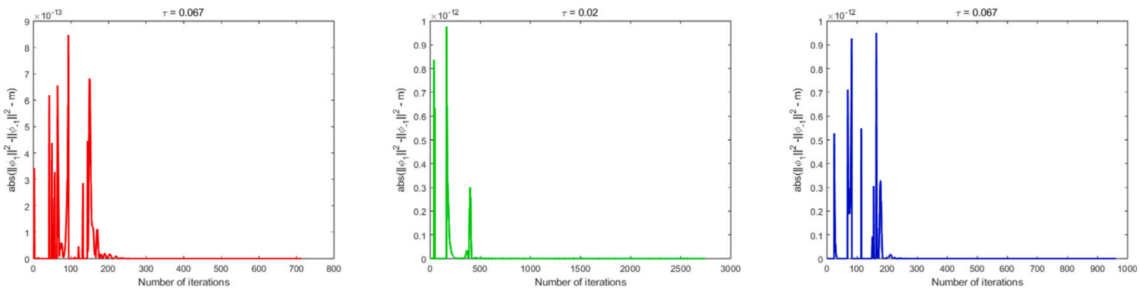

Fig. 6. Conservation of magnetization for Example 5.1 in Case II (Left: RCNAGF-SI; Middle: RCNAGF-LF; Right: RCNAGF-2SI).

θ is 0.1, and the damping parameter η is 3. From Figs. 5–7, we can see that our proposed schemes are able to maintain both mass and magnetization conservation well, and the magnitudes of all supplementary variables fall within the range of the system's compatibility error ϵ. The wave equations diagrams of Example 5.1 in two situations computed in the ground state in Fig. 8 are obtained using RCANGF-SI with a time step of $\tau = 1/15$, a stability parameter of $\theta = 0.05$, and a damping parameter of $\eta = 3$. The results are in line with the findings of the paper [15].

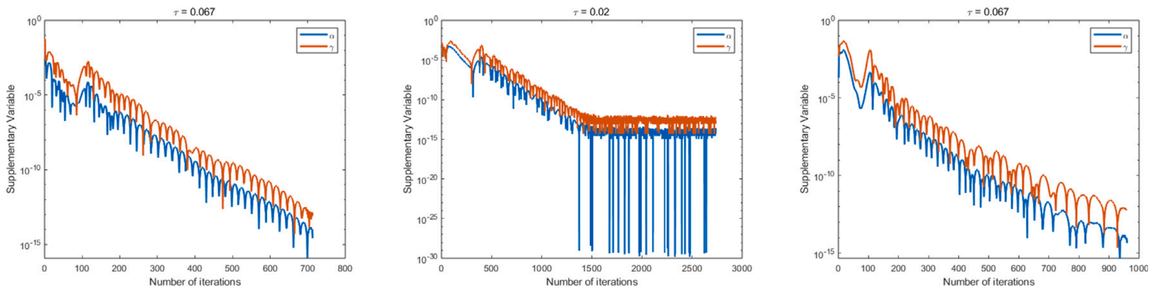

Fig. 7. Relationship between supplementary variables and number of iterations for Example 5.1 in Case II (Left: RCNAGF-SI; Middle: RCNAGF-LF; Right: RCNAGF-2SI).

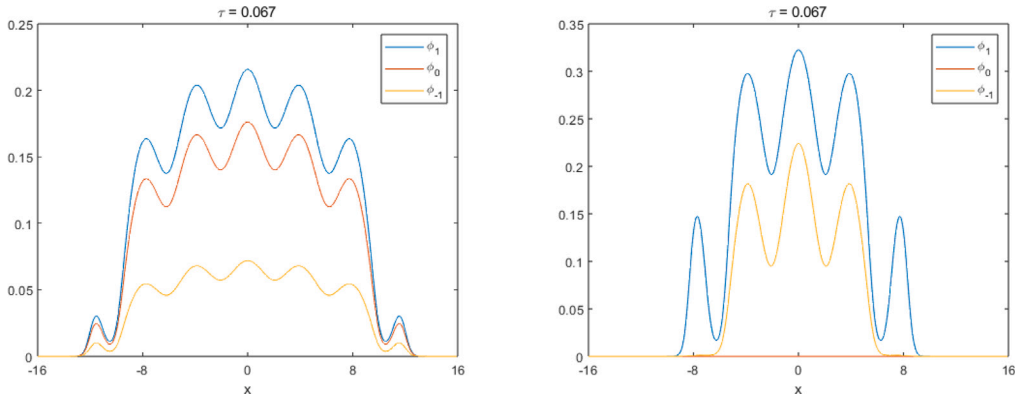

Fig. 8. Wave functions of the ground state in Example 5.1. Left: Case I and Right Case II.

Table 3
Ground state energy $E(\mathbf{\Phi}_g)$, Lagrange multipliers ξ_g and σ_g, and the maximum residual for the Euler–Lagrange equations "maxres" in Example 5.2.

Case	time step	$E(\mathbf{\Phi}_g)$	ξ_g	σ_g	maxre	θ	η
I	0.1	46.9809414676	52.2524304319	0.15024417676	7.0073E−10^{-7}	0.1	3
II	0.1	55.4362448029	63.36355828686	−0.0000244883	2.3101E−10^{-7}	0.1	30

5.2. Applications to spin-1 BECs in 3D

In this subsection, we report numerical results obtained using the RCNGF-SI scheme for the ground state solution of spin-1 BECs in 3D.

Example 5.2. We study ground states of 3D spin-1 BECs by the scheme RCNAGF-SI in an optical lattice potential $V(\mathbf{x}) = \sum_{j=x,y,z}[\frac{j^2}{2} + 100\sin^2(\frac{\pi j}{2})]$ with two different types of interactions.
Case I. (antiferromagnetic condensates) i.e. $\beta_n = 239$ and $\beta_s = 7.5$.
Case II. (ferromagnetic condensates) i.e. $\beta_n = 880$ and $\beta_s = -4.1$.

In our numerical experiments, we choose a time step of $\tau = 0.1$, a stability parameter of $\theta = 0.1$ and fixed magnetization $m = 0.5$ and the computational domain is set as $\Omega = [-8,8]\times[-8,8]\times[-8,8]$. We employ spatial Fourier-pseudo spectral discretization with a discrete Fourier space dimension of $Nf = 128\times128\times128$. We choose the initial data

$$\boldsymbol{\Phi}_I = \left(\sqrt{\frac{1+m}{2}}, 0, \sqrt{\frac{1-m}{2}}\right)^T \phi_g^{\mathrm{ap}}(\mathbf{x}) \quad \text{and} \quad \boldsymbol{\Phi}_{II} = \left(\frac{\sqrt{1+3m}}{2}, \sqrt{\frac{1-m}{2}}, \frac{\sqrt{1-m}}{2},\right)^T \phi_g^{\mathrm{ap}}(\mathbf{x})$$

with $\phi_g^{\mathrm{ap}}(\mathbf{x}) = e^{-(x^2+y^2+z^2)/2}/\pi^{3/4}$ for the Case I and Case II, respectively. The ground state is attained when the condition is met with a tolerance level of $\varepsilon = 10^{-7}$.

Based on the numerical results presented in Table 3 and Figs. 9 to 12, our results align with the findings presented in Paper [15]. It shows that our scheme is capable of computing the ground state solution for spin-1 BECs in 3D.

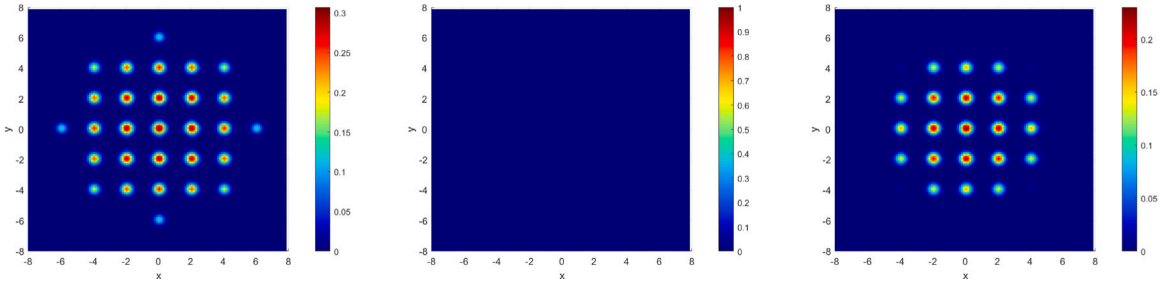

Fig. 9. Contour plots for the wave functions of the ground state of a spin-1 BEC in Example 5.2 for Case I, $\phi_1^g(x,y,0)$ (left), $\phi_0^g(x,y,0)$ (middle), $\phi_{-1}^g(x,y,0)$ (right).

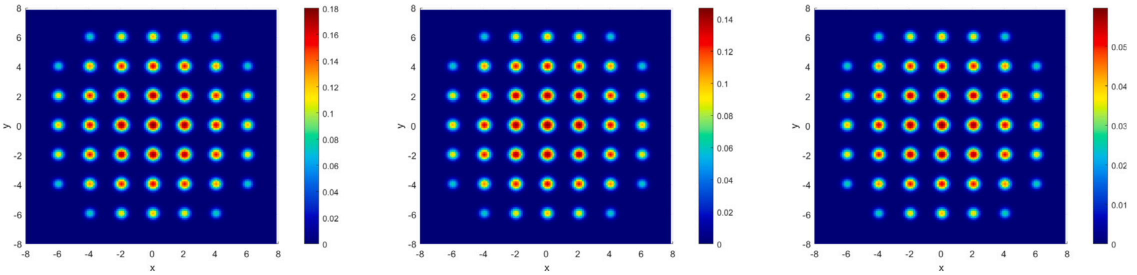

Fig. 10. Contour plots for the wave functions of the ground state of a spin-1 BEC in Example 5.2 for Case II, $\phi_1^g(x,y,0)$ (left), $\phi_0^g(x,y,0)$ (middle), $\phi_{-1}^g(x,y,0)$ (right).

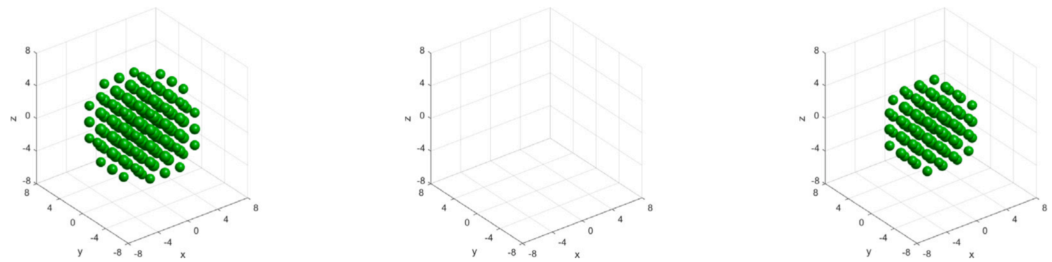

Fig. 11. Isosurface plots for the wave functions of the ground state of a spin-1 BEC in Example 5.2 for Case I, $\phi_1^g(x,y,z)=0.01$ (left), $\phi_0^g(x,y,z)=0.01$ (middle), $\phi_{-1}^g(x,y,z)=0.01$ (right).

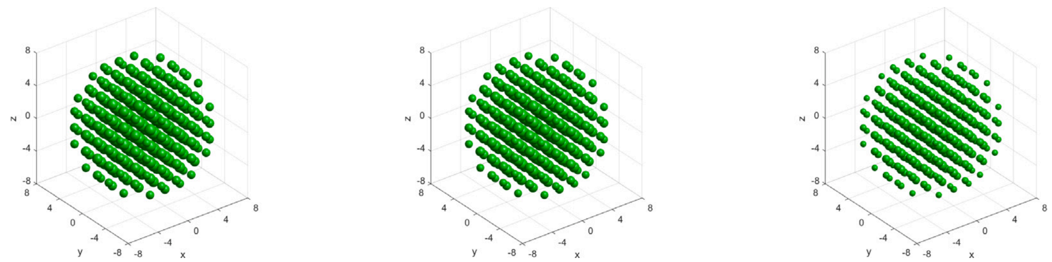

Fig. 12. Isosurface plots for the wave functions of the ground state of a spin-1 BEC in Example 5.2 for Case II, $\phi_1^g(x,y,z)=0.01$ (left), $\phi_0^g(x,y,z)=0.01$ (middle), $\phi_{-1}^g(x,y,z)=0.01$ (right).

6. Conclusion

In this paper, we introduce a continuous normalized accelerated gradient flow (CNAGF) to simulate the ground state of spin-1 BECs. By solving the steady-state solution while conserving mass and magnetization in CNAGF, we can accurately compute the ground states of spin-1 BECs. Since the Lagrange multiplier terms in CNAGF involve integrals, it is challenging to ensure that the solution satisfies the conservation of mass and magnetization constraints. Therefore, we constructed the reconfigured continuous normalized accelerated gradient flow (RCNAGF) inspired by the supplementary variable method, so that the solution of RCNAGF naturally satisfies these constraints. Some explicit and semi-implicit schemes are proposed to solve the steady-state solution of the RCNAGF for computing the ground states of spin-1 BECs. By selecting appropriate supplied functions, our schemes match the Euler–Lagrange equations upon termination of evolution. These results show that our proposed schemes can accurately compute the ground states of spin-1 BECs. The numerical experiment results show that the RCNAGF method is more efficient than the GFLM and RCNGF methods. Specifically, with the same time step size, the semi-implicit schemes RCNAGF-SI and RCNAGF-2SI require fewer iterations to meet the stopping criterion compared to GFLM-BF and RCNGF-BF. Additionally, the explicit scheme RCNAGF-LF allows for a larger time step size than GFLM-FE. The residuals of our numerical solutions for the Euler–Lagrange equations are roughly the same magnitude as the tolerance, indicating that our schemes can compute the ground states of BECs with high precision.

CRediT authorship contribution statement

Hao Wang: Software, Writing – review & editing, Methodology, Writing – original draft, Investigation. **Wenjun Cai:** Funding acquisition, Validation, Supervision. **Yongzhong Song:** Supervision. **Yushun Wang:** Validation, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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